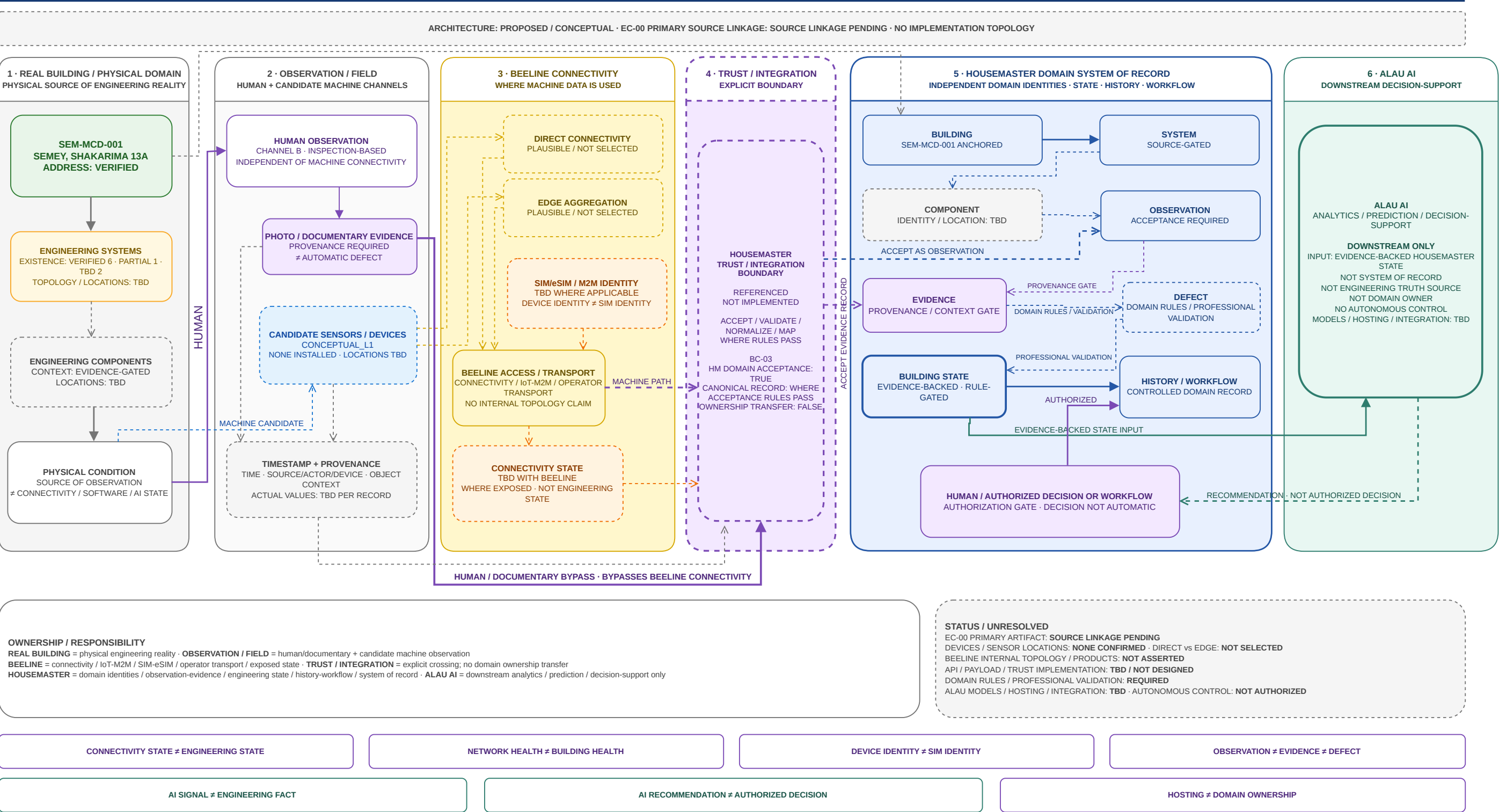


A3-EC00-01 · REFERENCE ARCHITECTURE OVERVIEW

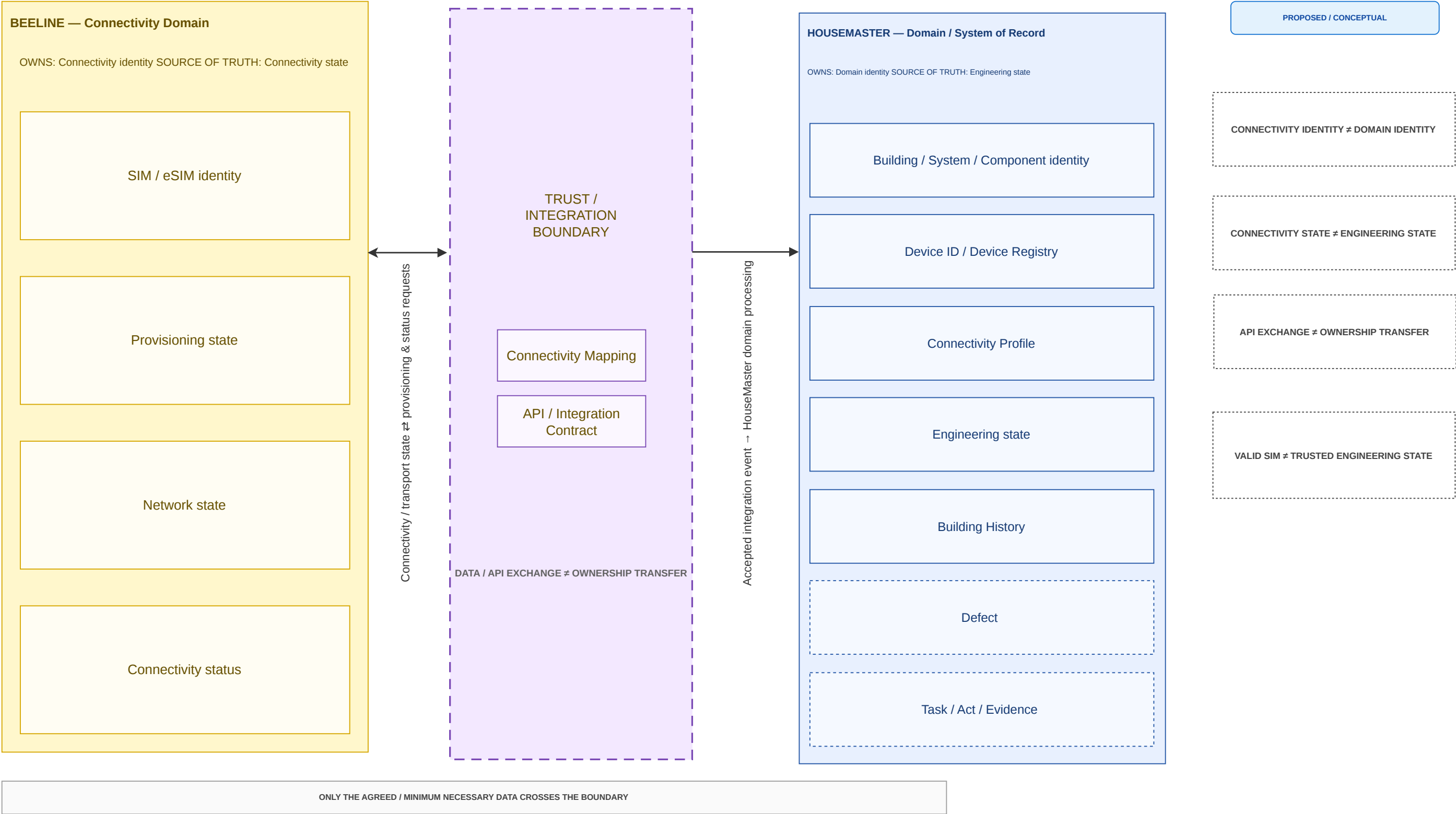
P2-01 · ASSEMBLY 001 · IR FROZEN

SEM-MCD-001 is physical reality · Beeline serves machine connectivity · HouseMaster remains the domain system of record · ALAU AI is downstream decision-support.



System Responsibility Model

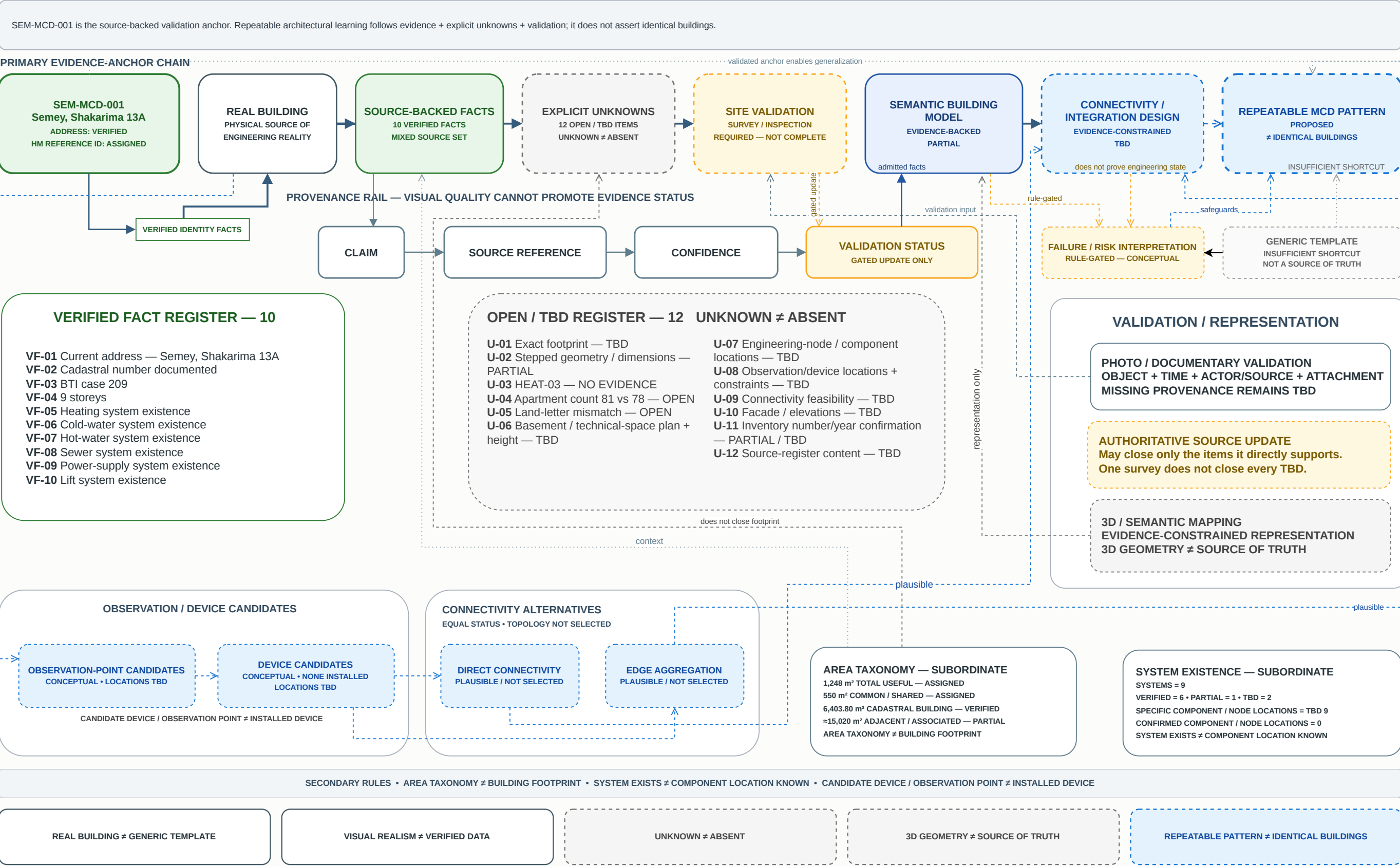
Beeline owns connectivity identity and connectivity state; HouseMaster owns domain identity and engineering state; integration connects the domains without transferring ownership.



Non-negotiable separations for connectivity, identity, evidence, engineering truth, authority and domain ownership.

REFERENCE: SEM-MCD-001 — SEMEY, SHAKARIMA 13A · AXIOMS: PROPOSED / CONCEPTUAL · EC-00 PRIMARY SOURCE LINKAGE: SOURCE LINKAGE PENDING · DUPLICATES: 0			
AXIOM	SEPARATION / RULE	WHAT IT PREVENTS	OPERATIONAL CONSEQUENCE
AX-01 A · STATE	CONNECTIVITY STATE ≠ ENGINEERING STATE	Reachability, degradation or restoration being treated as component/building condition.	Store connectivity state separately; never open, close or change engineering findings from connectivity alone.
AX-02 A · STATE	NETWORK HEALTH ≠ BUILDING HEALTH	Declaring a building healthy from network health or failed from network unavailability.	Use network telemetry only to qualify evidence availability; never substitute it for engineering evidence.
AX-08 A · STATE	DEVICE OFFLINE ≠ COMPONENT FAILED	False component-failure inference from missing telemetry or device silence.	Record an evidence gap; retain last confirmed state where applicable; investigate root cause before defect classification. · EC-03 §22
AX-03 B · IDENTITY	DEVICE IDENTITY ≠ SIM IDENTITY	HouseMaster domain identity/history changing with SIM replacement or provider lifecycle.	Maintain separate identifiers and explicit mappings; never derive HouseMaster IDs from SIM identifiers.
AX-04 C · TRUTH	OBSERVATION ≠ EVIDENCE ≠ DEFECT	Raw signal, note, photo or accepted record being treated as a verified defect.	Preserve separate records and explicit acceptance, provenance and professional-validation gates.
AX-05 C · TRUTH	AI SIGNAL ≠ ENGINEERING FACT	Model output overwriting evidence-backed HouseMaster state or becoming engineering truth.	Store AI signals separately; require evidence/domain validation before any engineering-state change.
AX-09 C · TRUTH	HUMAN EVIDENCE ≠ MACHINE DEPENDENCE	Treating all evidence as IoT-generated or blocking human inspection without machine data.	Maintain a direct human/documentary path with the same provenance and HouseMaster acceptance discipline. · EC-02 §4a.3
AX-06 D · AUTHORITY	AI RECOMMENDATION ≠ AUTHORIZED DECISION	Autonomous action or workflow-state change from an AI recommendation alone.	Route recommendations through explicit human/authorized review and record the accountable decision.
AX-07 D · OWNERSHIP	HOSTING ≠ DOMAIN OWNERSHIP	A provider being treated as owner of HouseMaster identities, evidence, state, history or workflow.	Preserve HouseMaster system-of-record ownership in contracts, integrations and responsibility matrices.
AX-10 D · OWNERSHIP	HOUSEMASTER DOMAIN ACCEPTANCE ≠ OWNERSHIP TRANSFER	Integration crossing being interpreted as external ownership of HouseMaster domain records.	HouseMaster accepts/validates/maps input and creates a canonical record where rules pass; ownership_transfer remains false. · OWNER-APPROVED BC-03
REAL BUILDING Physical source of engineering reality. Physical condition ≠ connectivity / software / AI state.			
BEELINE Connectivity / IoT-M2M / SIM-eSIM / operator transport / exposed state. Not domain owner; not building-condition authority.			
HOUSEMASTER Independent system of record: identities · observation/evidence · engineering state · history/workflow. Accepts / validates / maps at boundary.			
ALAU AI Downstream intelligence / prediction / decision-support. Not system of record · not engineering truth · no autonomous control.			
HUMAN / DOCUMENTARY CHANNEL Evidence-bearing input without Beeline connectivity. Provenance required · not IoT-dependent.			
AX-10 OPERATIONAL NOTE · OWNER-APPROVED FROZEN BC-03 housemaster_domain_acceptance = true · canonical_domain_record_created_where_acceptance_rules_pass = true · ownership_transfer = false			
FROZEN AXIOM REGISTER · 10 AXIOMS · 4 GROUPS · 5 DOMAIN CONSEQUENCES · DUPLICATE CANONICAL STATEMENTS: 0			

A3-EC01-01 REFERENCE BUILDING / EVIDENCE ANCHOR



SEM-MCD-001 Identity & Geometry Fact Sheet

SEM-MCD-001 has a documented building identity and basic geometric facts, while exact footprint geometry remains evidence-limited.

IDENTITY

REFERENCE ID (internal, not an evidentiary fact)

SEM-MCD-001

Location

г. Семей, пр. Шакарима, 13 «А»

VERIFIED

Cadastral number

05.252.025.162/543:13a

VERIFIED

BTI case

209

VERIFIED

Floors

9

VERIFIED

Entrances

3

DERIVED

A. IDENTITY

SOURCE

EC-01 §3

Location, cadastral number, BTI case, floor count

WHAT IT SUPPORTS

Specific documented building identity

WHAT IT DOES NOT SUPPORT

Complete building geometry

B. BASIC GEOMETRY

SOURCE

EC-01 §3/§4 via frozen source extract

WHAT IT SHOWS

9 floors; 3 entrances; stepped/non-rectangular footprint character

WHAT IT SUPPORTS

Basic geometry and building organization

WHAT IT DOES NOT SUPPORT

Precise footprint dimensions or area

CONFIDENCE

Mixed: VERIFIED / DERIVED / PARTIAL

C. UNRESOLVED GEOMETRY

WHAT IT SHOWS

Evidence gap

WHAT IT SUPPORTS

Explicitly unresolved exact footprint geometry

WHAT IT DOES NOT SUPPORT

Exact area; exact dimensions

CONFIDENCE

TBD

TBD

Exact footprint area; precise footprint dimensions

DOCUMENTED IDENTITY ≠ COMPLETE GEOMETRY

FOOTPRINT SHAPE ≠ EXACT DIMENSIONS

VISUAL REALISM ≠ EVIDENCE CERTAINTY

PARTIAL ≠ VERIFIED

PARTIAL / SCHEMATIC



Stepped / non-rectangular footprint character only — no dimension, scale, or area implied.

EXACT FOOTPRINT AREA: TBD

PRECISE FOOTPRINT DIMENSIONS: TBD

A3-EC01-03 AREA TAXONOMY

REFERENCE SEM-MCD-001 — Semey, Shakarima 13A

THESE FOUR FIGURES ≠ FOUR COMPETING MEASUREMENTS OF ONE AREA

They describe different semantic categories. No arithmetic hierarchy, footprint inference, or site-boundary inference is permitted.

PRIMARY VISUAL — AREA TAXONOMY MATRIX

CATEGORY	VALUE	SEMANTIC MEANING	SOURCE / STATUS	WHAT IT DOES NOT MEAN
AREA-01 TOTAL USEFUL BUILDING AREA	1,248 m² STATUS: ASSIGNED	Total useful area of the building under the approved taxonomy.	OWNER-APPROVED TAXONOMY ASSIGNED EC-01 §0b / §5.0 / BF-026 NOT INDEPENDENTLY RE-DERIVED	• Building footprint • Common area only • Land parcel area
AREA-02 COMMON / SHARED BUILDING AREA	550 m² STATUS: ASSIGNED	Entrances, stair landings and flights, lift shafts, technical rooms and basement under the approved taxonomy.	OWNER-APPROVED TAXONOMY ASSIGNED EC-01 §0b / §5.0 / BF-027 NOT INDEPENDENTLY RE-DERIVED	• Total useful building area • Exact basement area • Building footprint
AREA-03 CADASTRAL BUILDING AREA	6,403.8 m² FIGURE: VERIFIED	Cadastral building-area category associated with the building record.	SRC-006 FIGURE: VERIFIED CATEGORY: OWNER-APPROVED TAXONOMY FOOTPRINT NOT VALIDATED	• Exact footprint • Adjacent territory • Total useful area
AREA-04 ADJACENT / ASSOCIATED TERRITORY	≈15,020 m² STATUS: PARTIAL	Adjacent / associated territory category linked to the building context.	SRC-001/002 PARTIAL CATEGORY: OWNER-APPROVED TAXONOMY LEGAL BOUNDARY: NOT ESTABLISHED	• Building area • Building footprint • Confirmed legal parcel boundary

PROVENANCE RAIL



ASSIGNED ≠ PARTIAL ≠ VERIFIED • OWNER-APPROVED CATEGORY DOES NOT PROMOTE SOURCE CONFIDENCE

NON-EQUIVALENCE MODEL

TOTAL USEFUL AREA ≠ COMMON / SHARED AREA
CADASTRAL BUILDING AREA ≠ ADJACENT / ASSOCIATED TERRITORY
ALL FOUR AREA CATEGORIES ≠ EXACT BUILDING FOOTPRINT
ANY AREA VALUE ≠ VERIFIED GEOMETRY

PROHIBITED EQUATIONS
1,248 + 550 ≠ 6,403.8
6,403.8 + ... ≠ 15,020
NO RATIOS • NO SUMMATION • NO INFERRED DECOMPOSITION

FOOTPRINT / GEOMETRY OPEN ITEMS

EXACT BUILDING FOOTPRINT AREA
NOT ESTABLISHED / TBD

STEPPED / NON-RECTANGULAR
GEOMETRY
PARTIAL

AUTHORITATIVE
GEOMETRY / SITE
SURVEY
REQUIRED

6,403.8 m² vs 15,020 m² — CONFLICT: NO



Engineering Systems — Existence vs Location

Engineering systems are assessed separately at system-existence and component-location levels; confirmed system existence does not imply known component locations.

SYSTEM EXISTENCE · VERIFIED 6 / PARTIAL 1 / TBD 2			≠	COMPONENT LOCATION KNOWN · 0 CONFIRMED / 9 TBD	
SYSTEM	EXISTENCE STATUS	SOURCE / EVIDENCE · PROVENANCE / CONFIDENCE	LOCATION STATUS INDEPENDENT	WHAT IS KNOWN	WHAT REMAINS TBD
Отопление (Heating)	VERIFIED	Marked row on SRC-004/Image 5 checklist; corroborated by SRC-001/002 Primary checklist evidence; directly documented existence	TBD	The heating system exists	Topology and all component/node locations
ХВС (Cold water)	VERIFIED	Marked row on SRC-004/Image 5 checklist Primary checklist evidence; directly documented existence	TBD	The cold-water system exists	Topology and all component/node locations
ГВС (Hot water)	VERIFIED	Marked row on SRC-004/Image 5 checklist as Горячее водоснабжение Primary checklist evidence; directly documented existence	TBD	The hot-water system exists	Topology and all component/node locations
Канализация (Sewer)	VERIFIED	Marked row on SRC-004/Image 5 checklist Primary checklist evidence; directly documented existence	TBD	The sewer system exists	Topology and all component/node locations
Электроснабжение (Power)	VERIFIED	Marked row on SRC-004/Image 5 checklist Primary checklist evidence; directly documented existence	TBD	The electrical system exists	Topology and all component/node locations
Вентиляция (Ventilation)	TBD	Checklist row present on SRC-004/Image 5; mark legibility uncertain Primary source present but illegible for existence confirmation	TBD	A ventilation row is present in the checklist	Whether the system exists; topology; all component/node locations
Лифты (Lifts)	VERIFIED	Marked row on SRC-004/Image 5; SRC-001/002 claim 3 lifts, one per entrance Existence verified by primary checklist; count/association partial from secondary sources	TBD	A lift system exists; count and one-per-entrance association are PARTIAL	Topology and all specific component/node locations
Газоснабжение (Gas)	TBD	SRC-004/Image 5 checklist marks uncertain; SRC-001/002 claim gas is absent Primary mark illegible; absence claim secondary and unconfirmed	TBD	Checklist rows exist and secondary documents claim absence	Whether gas supply exists; topology; all component/node locations
Кровельный водосток (Roof drainage)	PARTIAL	Internal drainage claimed only by SRC-001/002; SRC-004/Image 5 contains a roof field Secondary-only system claim; partial confidence	TBD	An internal roof-drainage system is claimed	Primary confirmation; topology; all component/node locations
DOCUMENTED SYSTEM ≠ DOCUMENTED COMPONENT LOCATION			SYSTEM EXISTENCE ≠ AS-BUILT ENGINEERING SCHEME		
VISUAL SCHEMATIC ≠ FIELD-VERIFIED LOCATION			TBD LOCATION ≠ SYSTEM ABSENT		
SYSTEMS: 9 · EXISTENCE: VERIFIED 6 / PARTIAL 1 / TBD 2 · LOCATION: TBD 9 · CONFIRMED COMPONENT/NODE LOCATIONS: 0					

OBSERVABILITY NARRATIVE

HouseMaster starts with the engineering question that requires evidence — not with a sensor.

OBSERVABILITY ≠ SENSORIZATION

OBSERVABILITY ≠ FULL SENSOR COVERAGE

01 · PHYSICAL REALITY / ENGINEERING QUESTION

SEM-MCD-001 · REAL BUILDING

Semey, Shakarima 13A · evidence anchor

SYSTEM / COMPONENT / ZONE CONTEXT

Known context stays separate from unknown location.

OBSERVABLE CONDITION

physical state · value · change · event

OBSERVATION POINT

CANDIDATE / L1 · exact physical location = TBD

OBSERVATION POINT = PHYSICAL / ENGINEERING CONCEPT

Attached to building / system / component / zone context. May be served by Human, Machine, or both. It does not depend on an installed device.

OBSERVATION POINT ≠ DEVICE

PHYSICAL CONDITION ≠ OBSERVATION

OBS-CAND-001 · HEATING TEMPERATURE CONDITION

SYSTEM EXISTENCE: **VERIFIED** · OP: OC-001 / CANDIDATE L1
HUMAN: YES — IN PRINCIPLE · MACHINE: YES — PENDING FIELD SITING
DEVICE: CONCEPTUAL / NOT SELECTED / NOT INSTALLED · LOCATION: TBD

02 · TWO CO-APPLICABLE OBSERVATION CHANNELS

CHANNEL A · MACHINE / DEVICE-ASSISTED

STATUS: CONCEPTUAL / CANDIDATE

MAY SUPPORT: MEASURE · TIMESTAMP · BUFFER · FORWARD

DOES NOT MEAN: installed device · selected sensor · mounting point · protocol · coverage

CHANNEL B · HUMAN / DOCUMENTARY

STATUS: APPLICABLE WHERE ACCESS + CONTEXT ALLOW

INSPECTION · CHECKLIST · MANUAL READING · FIELD ASSESSMENT
PHOTO · NOTE · ACT · DOCUMENTARY RECORD

Independent of machine deployment; provenance remains required.

CO-APPLICABLE · NOT SEQUENTIAL · NOT MUTUALLY EXCLUSIVE

HUMAN OBSERVATION ≠ MACHINE DEPENDENCE
DEVICE CANDIDATE ≠ INSTALLED DEVICE

Human and Machine have equal semantic weight. Channel choice answers the engineering question; it does not rank trust by default.

03 · HOUSEMASTER ACCEPTANCE / ENGINEERING MEANING

PROVENANCE-BEARING OBSERVATION

PROVENANCE / CONTEXT GATE

EVIDENCE · WHERE CRITERIA PASS

DOMAIN / PROFESSIONAL VALIDATION

RULE-GATED INTERPRETATION

DEFECT AND/OR BUILDING STATE

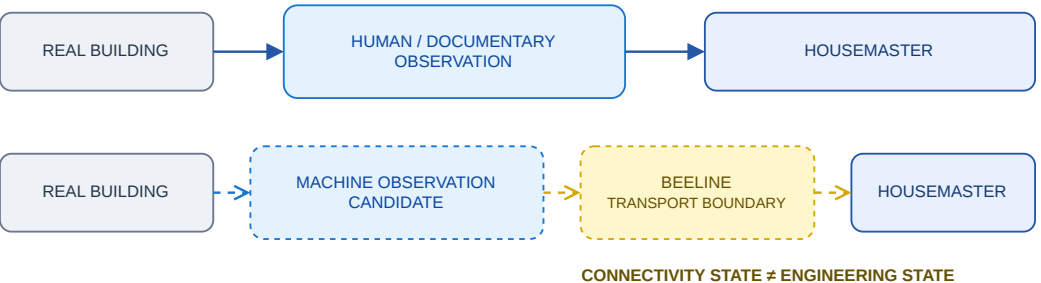
ONLY WHERE RULES PASS

OBSERVATION ≠ EVIDENCE · EVIDENCE ≠ DEFECT

HOUSEMASTER · SYSTEM OF RECORD

Anchors engineering context · preserves provenance · applies acceptance rules · maintains domain history/state

TWO-CHANNEL SUPPORTING MODEL



PROVENANCE / TIME

BUILDING / OBJECT CONTEXT
SYSTEM / COMPONENT / ZONE
OBSERVATION POINT
OBSERVATION CHANNEL
event_time
recorded / received time
ACTOR / SOURCE / DEVICE
ATTACHMENT / PHOTO
SOURCE PROVENANCE
CONFIDENCE / VALIDATION STATUS

event_time ≠ recorded / received time

NO EVENT ≠ NORMAL CONDITION

NO NEW OBSERVATION ≠
UNCHANGED ENGINEERING STATE

NO NEW OBSERVATION ≠ NO PHYSICAL
CHANGE

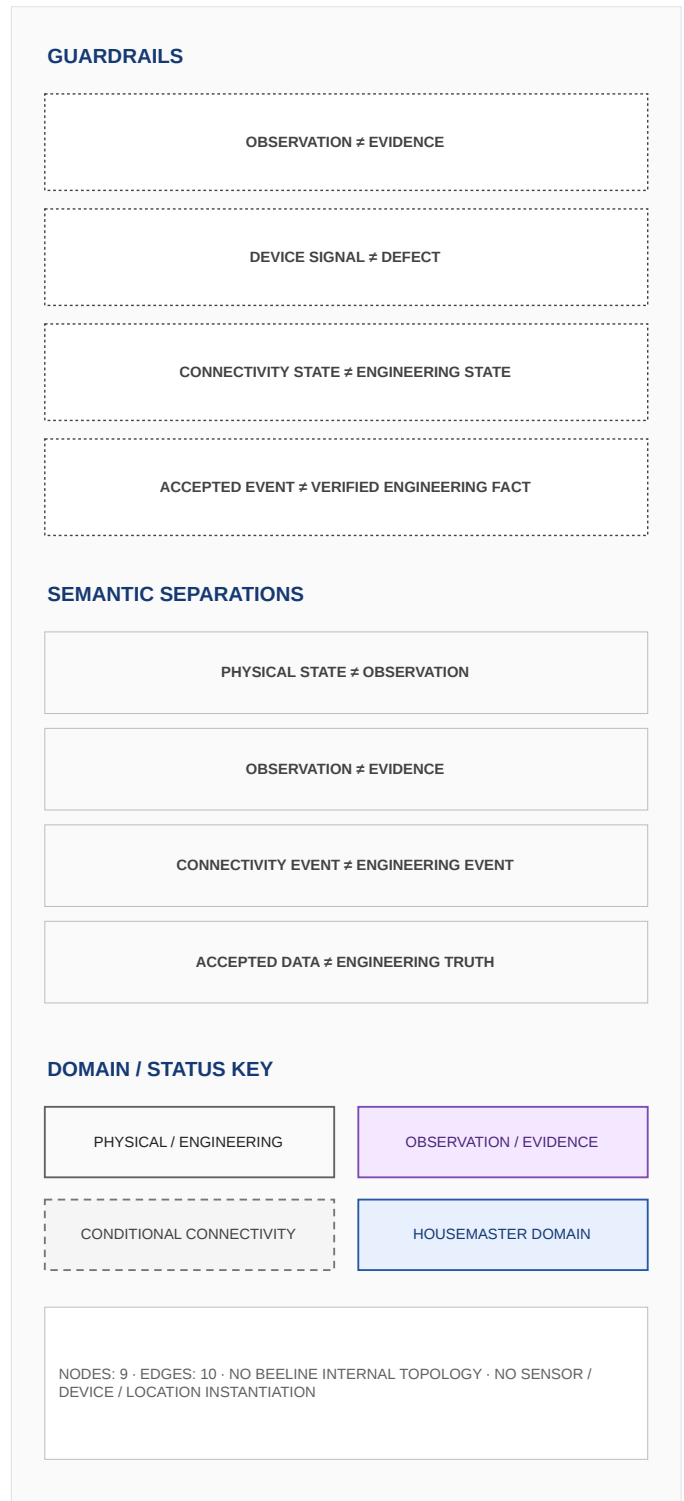
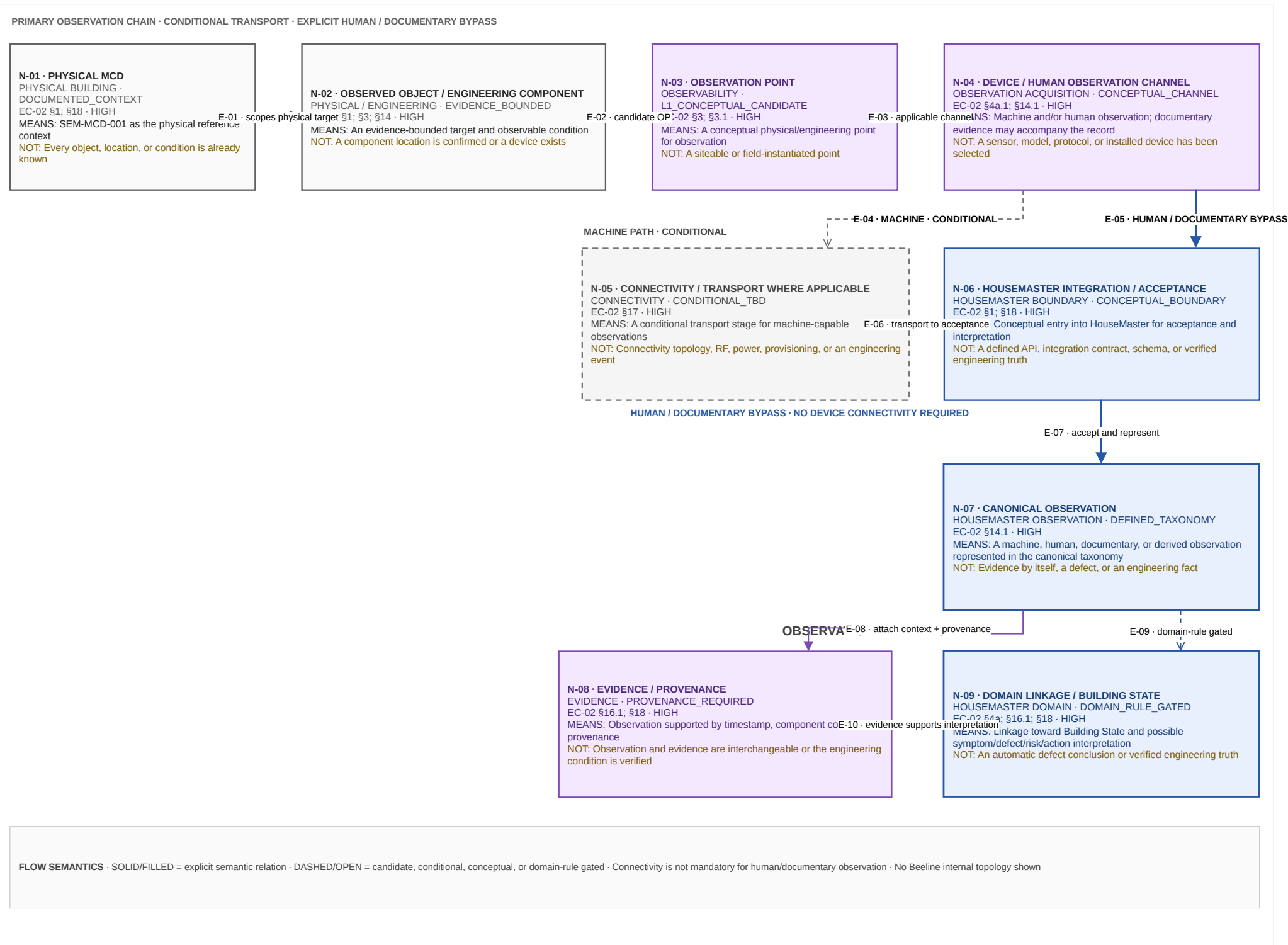
OPEN / TBD · 7 CATEGORIES

- 01 Observation-point physical locations — TBD
- 02 Device mounting locations — TBD
- 03 Installed-device status — TBD / NOT ESTABLISHED
- 04 Machine-observation coverage — TBD
- 05 Site validation — **REQUIRED** / NOT COMPLETED
- 06 Power / environmental / access constraints — TBD
- 07 Connectivity feasibility — TBD

NO COVERAGE % · NO SENSOR TARGET · NO MATURITY SCORE
VISUAL COVERAGE ≠ OBSERVABILITY COVERAGE

Hero Graph — Physical MCD to HouseMaster Observation

Show the engineering observation chain from the physical MCD to an accepted HouseMaster observation without conflating physical condition, connectivity, observation, evidence, or engineering state.



MASTER OBSERVABILITY MATRIX

SEM-MCD-001 · Semey, Shakarima 13A · rows 001–006

MASTER OBSERVABILITY MATRIX ≠ DEVICE DEPLOYMENT PLAN

MATRIX ROW COVERAGE ≠ FULL BUILDING OBSERVABILITY

OBSERVABILITY STATUS SUMMARY

ROWS 18 · CHANNELS 2 · BOTH 16 · DOCUMENTARY/HUMAN 2 · MACHINE ONLY 0
L1 18 · L2+ 0 · CONFIRMED OP LOCATIONS 0 · SITE VALIDATION 18 · INSTALLATION READY 0
INSTALLED DEVICES 0 CONFIRMED · ACTUAL PRESENCE NOT ESTABLISHED · 0 CONFIRMED ≠ PROVED ABSENCE
OBSERVABLE NOW THROUGH HUMAN 8 · AFTER FIELD MAPPING 10 · NO SCORE / % / MATURITY LADDER

CHANNEL / EVIDENCE LEGEND

HUMAN · MACHINE · BOTH · TBD
VERIFIED · PARTIAL · TBD · NO EVIDENCE · CONCEPTUAL / CANDIDATE
NOT SEVERITY · NOT RISK

ID	ENGINEERING CONTEXT	OBSERVABLE CONDITION / QUESTION	OP STATUS	HUMAN CHANNEL	MACHINE CHANNEL	EVIDENCE / PROVENANCE	VALIDATION / OPEN ITEMS	DOWNSTREAM RESTRICTION
OBS-CAND-001 OC-001 HEATING	SEM-MCD-001 / heating system entry / component location TBD	Temperature condition Is the heating temperature condition adequate and how does it vary over time?	L1 CONCEPTUAL TBD HEATING ENTRY LOCATION OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	YES General assessment / periodic inspection note	YES · CONCEPTUAL SCALAR temperature measurement / continuous-high-frequency concept CONN: REQUIRED / TBD / DOWNSTREAM	VERIFIED SYSTEM EXISTENCE LOCATION TBD EC-02 §15 OBS-CAND-001 / §15.x / §16; BF-015 as cited by EC-02 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE NOW HUMAN MACHINE PENDING SITING SITE: YES TBD-005 heating-entry location · component identity · machine siting · environment · power · access · connectivity feasibility · provenance completeness	DO NOT: DO NOT SITE OR DEPLOY MACHINE DEVICE UNTIL LOCATION AND SITE REQUIREMENTS ARE VALIDATED 0 confirmed ≠ proved absence
OBS-CAND-002 OC-001 HEATING	SEM-MCD-001 / heating system entry / component location TBD	Pressure condition What is the heating pressure condition at an accessible and validated point?	L1 CONCEPTUAL TBD OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	CONDITIONAL Manual gauge reading where an accessible gauge exists	YES · CONCEPTUAL SCALAR pressure measurement / continuous-high-frequency concept CONN: REQUIRED / TBD / DOWNSTREAM	VERIFIED SYSTEM EXISTENCE LOCATION TBD EC-02 §15 OBS-CAND-002 / §15.x; BF-015 as cited by EC-02 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE AFTER FIELD MAPPING SITE: YES TBD-005 pressure point · gauge existence · gauge access · exact location · machine siting · connectivity feasibility · provenance completeness	DO NOT: DO NOT ASSUME GAUGE OR MACHINE POINT EXISTS BEFORE FIELD MAPPING 0 confirmed ≠ proved absence
OBS-CAND-003 OC-002 HEATING	SEM-MCD-001 / heating metering node / node location and interface existence TBD	Flow/consumption metering condition Can heating flow or consumption be observed through a confirmed metering context?	L1 CONCEPTUAL TBD METERING NODE OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	YES Manual meter/interface reading	YES · CONCEPTUAL COUNTER / meter or pulse-interface reading / periodic concept CONN: REQUIRED / TBD / DOWNSTREAM	VERIFIED SYSTEM EXISTENCE METER INTERFACE TBD EC-02 §15 OBS-CAND-003 / §15.x; BF-015 as cited by EC-02 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE AFTER FIELD MAPPING SITE: YES TBD-005 metering node · interface existence · access · exact location · machine feasibility · connectivity feasibility · provenance completeness	DO NOT: DO NOT DESIGN INTERFACE OR DEPLOYMENT UNTIL METERING CONTEXT IS CONFIRMED 0 confirmed ≠ proved absence
OBS-CAND-004 OC-004 COLD WATER	SEM-MCD-001 / cold-water system entry / component location TBD	Pressure condition What is the cold-water pressure condition at a validated entry point?	L1 CONCEPTUAL TBD OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	CONDITIONAL Manual gauge reading where accessible	YES · CONCEPTUAL SCALAR pressure measurement / continuous-high-frequency concept CONN: REQUIRED / TBD / DOWNSTREAM	VERIFIED SYSTEM EXISTENCE LOCATION TBD EC-02 §15 OBS-CAND-004 / §15.x; BF-016 as cited by EC-02 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE AFTER FIELD MAPPING SITE: YES TBD-005 entry location · gauge existence · access · machine siting · connectivity feasibility · provenance completeness	DO NOT: DO NOT ASSUME OR SITE PRESSURE OBSERVATION POINT BEFORE FIELD MAPPING 0 confirmed ≠ proved absence
OBS-CAND-005 OC-004 COLD WATER	SEM-MCD-001 / cold-water system entry / component location TBD	Flow condition What is the cold-water flow condition at a validated entry point?	L1 CONCEPTUAL TBD OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	CONDITIONAL Manual/visual assessment where a readable flow point exists	YES · CONCEPTUAL SCALAR flow measurement / continuous-high-frequency concept CONN: REQUIRED / TBD / DOWNSTREAM	VERIFIED SYSTEM EXISTENCE LOCATION TBD EC-02 §15 OBS-CAND-005 / §15.x; BF-016 as cited by EC-02 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE AFTER FIELD MAPPING SITE: YES TBD-005 entry/flow point · access · machine siting · power/environment · connectivity feasibility · provenance completeness	DO NOT: DO NOT CREATE FLOW DEPLOYMENT PLAN BEFORE FIELD VALIDATION 0 confirmed ≠ proved absence
OBS-CAND-006 OC-005 COLD WATER	SEM-MCD-001 / cold-water metering node / node location and interface existence TBD	Flow/consumption metering condition Can cold-water flow or consumption be observed through a confirmed metering context?	L1 CONCEPTUAL TBD METERING NODE OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	YES Manual meter/interface reading	YES · CONCEPTUAL COUNTER / meter-interface reading / periodic concept CONN: REQUIRED / TBD / DOWNSTREAM	VERIFIED SYSTEM EXISTENCE METER INTERFACE TBD EC-02 §15 OBS-CAND-006 / §15.x; BF-016 as cited by EC-02 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE AFTER FIELD MAPPING SITE: YES TBD-005 metering node · interface existence · access · machine feasibility · connectivity feasibility · provenance completeness	DO NOT: DO NOT DEFINE INTERFACE OR CONNECTIVITY BEFORE METERING CONTEXT VALIDATION 0 confirmed ≠ proved absence

MASTER OBSERVABILITY MATRIX ≠ DEVICE DEPLOYMENT PLAN · OBSERVATION POINT ≠ DEVICE · CANDIDATE ≠ INSTALLED DEVICE · SYSTEM EXISTS ≠ COMPONENT LOCATION KNOWN · MATRIX ROW COVERAGE ≠ FULL BUILDING OBSERVABILITY

MASTER OBSERVABILITY MATRIX

SEM-MCD-001 · Semey, Shakarima 13A · rows 007–012

MASTER OBSERVABILITY MATRIX ≠ DEVICE DEPLOYMENT PLAN

MATRIX ROW COVERAGE ≠ FULL BUILDING OBSERVABILITY

TIME / EVIDENCE BOUNDARIES

event_time ≠ recorded / received time · NO EVENT ≠ NORMAL CONDITION · NO NEW OBSERVATION ≠ UNCHANGED ENGINEERING STATE
OBSERVATION ≠ EVIDENCE · EVIDENCE ≠ DEFECT · engineering meaning remains domain / professional-validation gated

CHANNEL RELATIONSHIP

HUMAN + MACHINE = CO-APPLICABLE · NOT SEQUENTIAL · NOT MUTUALLY EXCLUSIVE
OBS-CAND-014 = DOCUMENTARY/HUMAN · NO CURRENT MACHINE MODEL
OBS-CAND-016 = DOCUMENTARY/HUMAN · NO MACHINE TARGET

ID	ENGINEERING CONTEXT	OBSERVABLE CONDITION / QUESTION	OP STATUS	HUMAN CHANNEL	MACHINE CHANNEL	EVIDENCE / PROVENANCE	VALIDATION / OPEN ITEMS	DOWNSTREAM RESTRICTION
OBS-CAND-007 OC-006 HOT WATER	SEM-MCD-001 / hot-water system / location TBD / ontology gap retained	Temperature/availability condition Is hot water available at an adequate temperature in the observed context?	L1 CONCEPTUAL TBD OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	YES Tap-level availability/temperature check and note	YES · CONCEPTUAL SCALAR temperature or availability-state measurement / continuous-high-frequency concept CONN: REQUIRED / TBD / DOWNSTREAM	VERIFIED SYSTEM EXISTENCE LOCATION TBD ONTOLOGY GAP EC-02 §15 OBS-CAND-007 / §15.x / §7; BF-017 as cited by EC-02 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE NOW HUMAN MACHINE PENDING SITING SITE: YES TBD-005 location · ontology mapping · access · machine siting · connectivity feasibility · provenance completeness	DO NOT: DO NOT INSTANTIATE COMPONENT ID OR DEVICE BEFORE EVIDENCE AND FIELD VALIDATION 0 confirmed ≠ proved absence
OBS-CAND-008 OC-006 HOT WATER	SEM-MCD-001 / hot-water system / location TBD	Leakage/water-presence indication Is there a leakage or water-presence indication associated with the hot-water context?	L1 CONCEPTUAL TBD OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	YES Visual assessment / photo / note where accessible	YES · CONCEPTUAL EVENT or BINARY STATE capture / event-driven concept CONN: REQUIRED / TBD / DOWNSTREAM	VERIFIED SYSTEM EXISTENCE LOCATION TBD EC-02 §15 OBS-CAND-008 / §15.x; BF-017 as cited by EC-02 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE NOW HUMAN MACHINE PENDING SITING SITE: YES TBD-005 observation point · access · cause interpretation · machine siting · connectivity feasibility · provenance completeness	DO NOT: DO NOT CLASSIFY DEFECT OR SITE DEVICE FROM INDICATION ALONE 0 confirmed ≠ proved absence
OBS-CAND-009 OC-007 BASEMENT ENVIRONMENT	SEM-MCD-001 / basement zone / zone existence PARTIAL secondary-source only	Water-presence/flooding indication Is water presence or flooding observed in a field-confirmed basement environment?	WEAK / PARTIAL CONTEXT TBD ZONE AND POINT OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	YES Inspection / photo / location note	YES · CONCEPTUAL EVENT presence/absence capture / event-driven concept CONN: REQUIRED / TBD / DOWNSTREAM	PARTIAL ZONE EXISTENCE PRIMARY OR FIELD CONFIRMATION REQUIRED EC-02 §15 OBS-CAND-009 / §15.x / §12 TBD-EC02-04 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE AFTER FIELD MAPPING SITE: YES TBD-003 AND TBD-EC02-04 basement existence · access · point location · field confirmation · machine siting · power/environment · connectivity feasibility · provenance completeness	DO NOT: DO NOT PROMOTE BEYOND PROVENANCE GATED L1 UNTIL ZONE IS PRIMARY OR FIELD CONFIRMED 0 confirmed ≠ proved absence
OBS-CAND-010 OC-007 BASEMENT ENVIRONMENT	SEM-MCD-001 / basement zone / zone existence PARTIAL secondary-source only	Temperature condition What is the temperature condition in a field-confirmed basement environment?	WEAK / PARTIAL CONTEXT TBD ZONE AND POINT OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	YES Manual assessment/reading and note where accessible	YES · CONCEPTUAL SCALAR temperature measurement / periodic concept CONN: REQUIRED / TBD / DOWNSTREAM	PARTIAL ZONE EXISTENCE SITE: YES TBD-003 AND TBD-EC02-04 EC-02 §15 OBS-CAND-010 / §15.x / §12 TBD-EC02-04 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE AFTER FIELD MAPPING SITE: YES TBD-003 AND TBD-EC02-04 basement existence · access · point location · field confirmation · machine siting · power/environment · connectivity feasibility · provenance completeness	DO NOT: DO NOT SITE OR SCORE BASEMENT TEMPERATURE OBSERVABILITY BEFORE ZONE CONFIRMATION 0 confirmed ≠ proved absence
OBS-CAND-011 OC-007 BASEMENT ENVIRONMENT	SEM-MCD-001 / basement zone / zone existence PARTIAL secondary-source only	Humidity condition What is the humidity condition in a field-confirmed basement environment?	WEAK / PARTIAL CONTEXT TBD ZONE AND POINT OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	YES Manual assessment/reading and note where accessible	YES · CONCEPTUAL SCALAR humidity measurement / periodic concept CONN: REQUIRED / TBD / DOWNSTREAM	PARTIAL ZONE EXISTENCE SITE: YES TBD-003 AND TBD-EC02-04 EC-02 §15 OBS-CAND-011 / §15.x / §12 TBD-EC02-04 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE AFTER FIELD MAPPING SITE: YES TBD-003 AND TBD-EC02-04 basement existence · access · point location · field confirmation · machine siting · power/environment · connectivity feasibility · provenance completeness	DO NOT: DO NOT SITE OR SCORE BASEMENT HUMIDITY OBSERVABILITY BEFORE ZONE CONFIRMATION 0 confirmed ≠ proved absence
OBS-CAND-012 OC-008 ELECTRICAL	SEM-MCD-001 / main distribution board / location TBD	Supply-availability condition Is electrical supply available in the validated main-board context?	L1 CONCEPTUAL TBD MAIN BOARD OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	YES General availability assessment / inspection note	YES · CONCEPTUAL BINARY STATE capture / event-driven concept CONN: REQUIRED / TBD / DOWNSTREAM	VERIFIED SYSTEM EXISTENCE LOCATION TBD EC-02 §15 OBS-CAND-012 / §15.x; BF-019 as cited by EC-02 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE NOW HUMAN MACHINE PENDING SITING SITE: YES TBD-005 main-board location · access · observation point · machine siting · power/environment · connectivity feasibility · provenance completeness	DO NOT: DO NOT SITE MACHINE OBSERVATION OR INFER BOARD LOCATION BEFORE FIELD MAPPING 0 confirmed ≠ proved absence

MASTER OBSERVABILITY MATRIX ≠ DEVICE DEPLOYMENT PLAN · OBSERVATION POINT ≠ DEVICE · CANDIDATE ≠ INSTALLED DEVICE · SYSTEM EXISTS ≠ COMPONENT LOCATION KNOWN · MATRIX ROW COVERAGE ≠ FULL BUILDING OBSERVABILITY

MASTER OBSERVABILITY MATRIX

SEM-MCD-001 · Semey, Shakarima 13A · rows 013–018

MASTER OBSERVABILITY MATRIX ≠ DEVICE DEPLOYMENT PLAN

MATRIX ROW COVERAGE ≠ FULL BUILDING OBSERVABILITY

SYSTEM / LOCATION + OBSERVATION-POINT BOUNDARIES

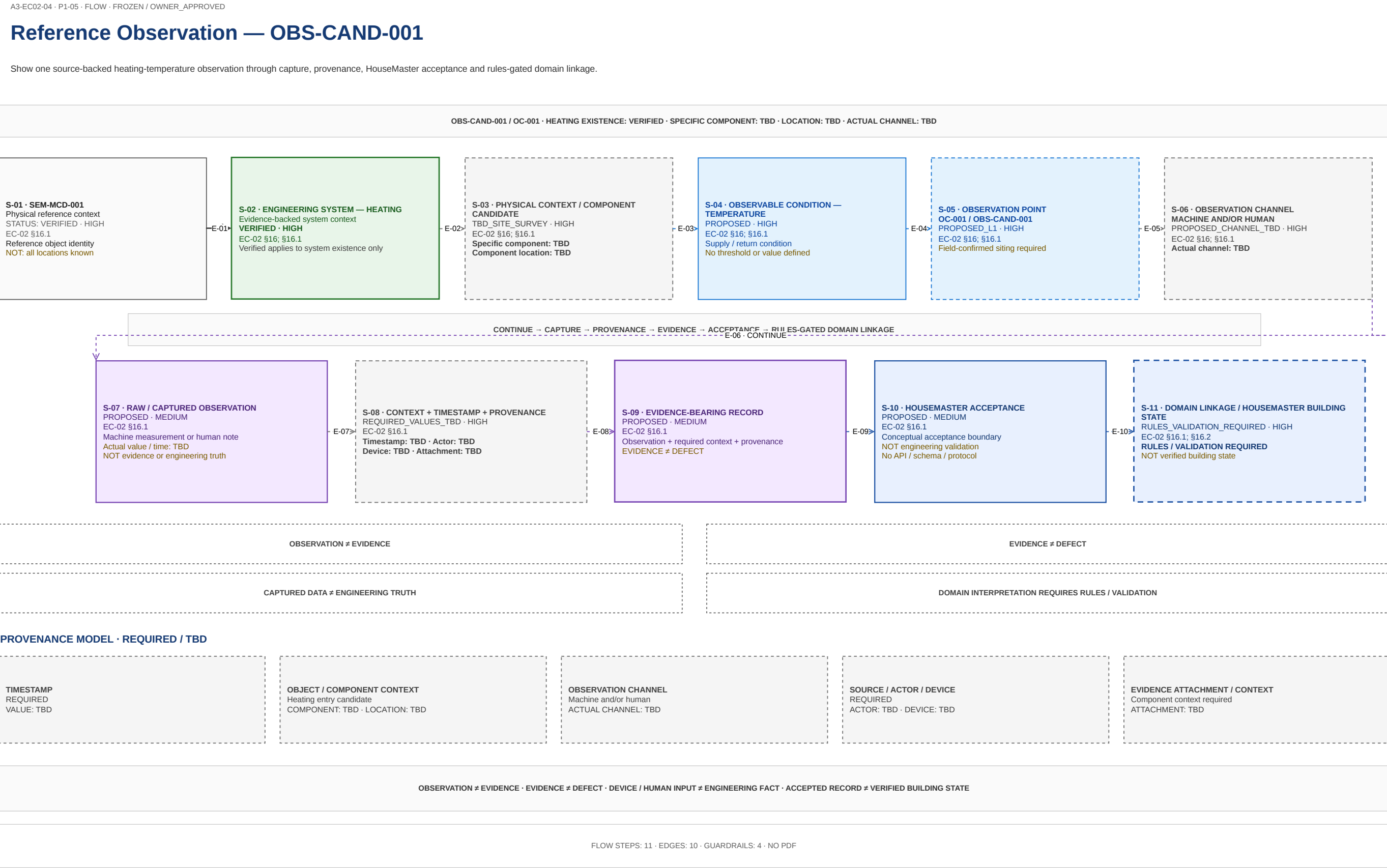
SYSTEM EXISTS ≠ COMPONENT LOCATION KNOWN · OBSERVATION POINT ≠ DEVICE · CANDIDATE ≠ INSTALLED DEVICE · TBD LOCATION ≠ NO PHYSICAL COMPONENT

COVERAGE / DEVICE SEMANTICS

MATRIX ROW COVERAGE ≠ FULL BUILDING OBSERVABILITY
INSTALLED DEVICES = 0 CONFIRMED · ACTUAL PRESENCE = NOT ESTABLISHED
0 CONFIRMED ≠ PROVED ABSENCE · NO SCORE / % / SENSOR TARGET / MATURITY

ID	ENGINEERING CONTEXT	OBSERVABLE CONDITION / QUESTION	OP STATUS	HUMAN CHANNEL	MACHINE CHANNEL	EVIDENCE / PROVENANCE	VALIDATION / OPEN ITEMS	DOWNSTREAM RESTRICTION
OBS-CAND-013 OC-008 ELECTRICAL	SEM-MCD-001 / main distribution board / location TBD	Outage/event condition Has an electrical outage/event occurred in the validated system context?	L1 · CONCEPTUAL TBD MAIN BOARD OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	YES Human report / inspection note / documentary record	YES · CONCEPTUAL EVENT capture / event-driven concept CONN: REQUIRED / TBD / DOWNSTREAM	VERIFIED SYSTEM EXISTENCE LOCATION TBD EC-02 §15 OBS-CAND-013 / §15.x; BF- 019 as cited by EC-02 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE NOW HUMAN MACHINE PENDING SITING SITE: YES TBD-005 main-board location · event context · access · machine siting · connectivity feasibility · provenance completeness	DO NOT: DO NOT EQUATE EVENT OR CONNECTIVITY LOSS WITH ENGINEERING DEFECT 0 confirmed ≠ proved absence
OBS-CAND-014 OC-009 SEWER	SEM-MCD-001 / sewer collector or outlet / location TBD / ontology OP undefined	Overflow/backflow indication Is an overflow or backflow indication visible in the validated sewer context?	L1 · ONTOLOGY GAP TBD COLLECTOR OUTLET OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	YES Visual assessment / inspection note / photo or documentary record	NO CURRENT MODEL NOT DEFINED IN CURRENT SOURCE MODEL CONN: NOT REQUIRED · HUMAN/DOC MODEL	VERIFIED SYSTEM EXISTENCE LOCATION TBD ONTOLOGY GAP EC-02 §15 OBS-CAND-014 / §15.x / §7; BF-018 as cited by EC-02 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE NOW HUMAN SITE: YES TBD-005 collector/outlet location · access · ontology OP extension · provenance completeness · professional interpretation	DO NOT: DO NOT CREATE MACHINE CONNECTIVITY OR DEVICE REQUIREMENT FOR THIS ROW UNDER CURRENT MODEL 0 confirmed ≠ proved absence
OBS-CAND-015 OC-010 ROOF	SEM-MCD-001 / roof surface / roof existence-type PARTIAL / access TBD	Moisture/water-ingress condition Is moisture or water ingress observed at an accessible and validated roof context?	WEAK / PARTIAL CONTEXT TBD ROOF ACCESS AND POINT OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	YES Inspection / visual assessment / photo / note	YES · CONCEPTUAL EVENT moisture/water-ingress capture / event-driven concept CONN: REQUIRED / TBD / DOWNSTREAM	PARTIAL ROOF EXISTENCE TYPE ACCESS TBD EC-02 §15 OBS-CAND-015 / §15.x; roof context PARTIAL as carried by EC- 02 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE AFTER FIELD MAPPING SITE: YES TBD-004 roof existence/type confirmation · access · point location · machine suitability · power/environment · connectivity feasibility · provenance completeness	DO NOT: DO NOT SITE OPTIONAL MACHINE OBSERVATION BEFORE ROOF ACCESS AND POINT VALIDATION 0 confirmed ≠ proved absence
OBS-CAND-016 OC-010 ROOF	SEM-MCD-001 / roof surface / roof existence-type PARTIAL / access TBD	Visual defect evidence Is there visual defect evidence in an accessible and validated roof context?	WEAK / PARTIAL CONTEXT TBD ROOF ACCESS AND CONTEXT OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	YES Photo / inspection / note / act or documentary record	NO MACHINE TARGET NOT APPLICABLE CONN: NOT REQUIRED · HUMAN/DOC MODEL	PARTIAL ROOF EXISTENCE TYPE ACCESS TBD EC-02 §15 OBS-CAND-016 / §15.x; roof context PARTIAL as carried by EC- 02 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE AFTER FIELD MAPPING SITE: YES TBD-004 roof confirmation · access · exact context · attachment provenance · professional validation	DO NOT: DO NOT CREATE MACHINE DEVICE OR CONNECTIVITY SCOPE FOR DOCUMENTARY ONLY ROW 0 confirmed ≠ proved absence
OBS-CAND-017 OC-011 LIFTS	SEM-MCD-001 / lift per entrance / count PARTIAL / per-unit location TBD	Availability/operational condition Is each validated lift available and operational?	L1 · CONCEPTUAL TBD PER ENTRANCE LIFT LOCATION OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	YES General availability assessment / inspection or maintenance note	YES · CONCEPTUAL BINARY STATE availability capture / event-driven concept CONN: REQUIRED / TBD / DOWNSTREAM	VERIFIED SYSTEM EXISTENCE COUNT PARTIAL PER UNIT LOCATION TBD EC-02 §15 OBS-CAND-017 / §15.x / §12 TBD-EC02-01; BF-020 as cited by EC-02 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE NOW HUMAN MACHINE PENDING SITING SITE: YES TBD-EC02-01 lift count confirmation · per-entrance locations · access · machine interface/siting · connectivity feasibility · provenance completeness	DO NOT: DO NOT CREATE PER LIFT DEPLOYMENT BEFORE COUNT AND LOCATION CONFIRMATION 0 confirmed ≠ proved absence
OBS-CAND-018 OC-011 LIFTS	SEM-MCD-001 / lift per entrance / count PARTIAL / per-unit location TBD	Outage/event condition Has a lift outage/event occurred in a validated per- unit context?	L1 · CONCEPTUAL TBD PER ENTRANCE LIFT LOCATION OP ≠ DEVICE · CANDIDATE ≠ INSTALLED	YES Human report / inspection or maintenance note	YES · CONCEPTUAL EVENT capture / event-driven concept CONN: REQUIRED / TBD / DOWNSTREAM	VERIFIED SYSTEM EXISTENCE COUNT PARTIAL PER UNIT LOCATION TBD EC-02 §15 OBS-CAND-018 / §15.x / §12 TBD-EC02-01; BF-020 as cited by EC-02 CTX · OP · CH · TIME · SRC/ACTOR · RECORD · CONF · VALID	OBSERVABLE NOW HUMAN MACHINE PENDING SITING SITE: YES TBD-EC02-01 lift count confirmation · per-entrance locations · event context · machine interface/siting · connectivity feasibility · provenance completeness	DO NOT: DO NOT EQUATE EVENT OR CHANNEL LOSS WITH LIFT ENGINEERING DEFECT 0 confirmed ≠ proved absence

MASTER OBSERVABILITY MATRIX ≠ DEVICE DEPLOYMENT PLAN · OBSERVATION POINT ≠ DEVICE · CANDIDATE ≠ INSTALLED DEVICE · SYSTEM EXISTS ≠ COMPONENT LOCATION KNOWN · MATRIX ROW COVERAGE ≠ FULL BUILDING OBSERVABILITY



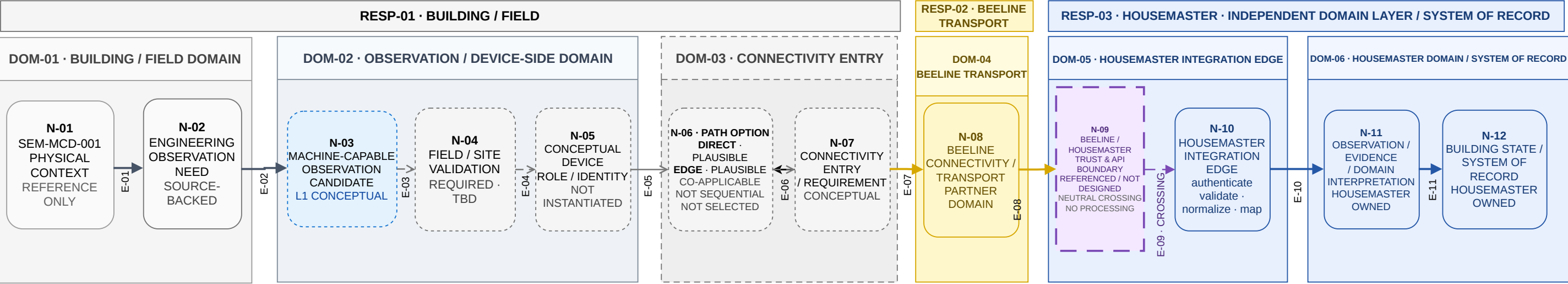
A3-EC03-01

REFERENCE · SEM-MCD-001
Semey, Shakarima 13A
Deployment: NOT ESTABLISHED ·
Installed devices: 0 CONFIRMED

CONNECTIVITY ENTRY ARCHITECTURE

Connectivity serves an observation requirement; it does not define the engineering meaning of the observation.

FROZEN SEMANTICS
DRAWIO REVIEW



THREE ARCHITECTURAL PLANES

DATA PLANE
Readings / states / scalar / event / counter
Preserve original **event_time**

CONTROL PLANE · NOT ENABLED BY DEFAULT
Remote actuation out of scope.

OPERATIONS PLANE
Connectivity status · reachability · lifecycle · diagnostics
Exact Beeline exposure: TBD

MINIMUM INTEGRATION SURFACE · CONCEPTUAL

- 1 · Observation/event content
scalar · event · state · counter
- 2 · Source/device + connectivity-profile identity references
TBD until deployment
- 3 · Connectivity / delivery status
exposed mapping **TBD WITH BEELINE**
- 4 · Temporal provenance · **event_time** ≠ **received_time**
- 5 · Integration outcome
validate · normalize · map · no payload schema
HouseMaster responsibility

OPEN / TBD REGISTER · 20 UNRESOLVED

SITE / DEVICE FEASIBILITY — 5
TBD-EC03-01 · power availability
TBD-EC03-02 · physical access
TBD-EC03-03 · RF/connectivity
environment
TBD-EC03-04 · Direct vs Edge
preference
TBD-EC03-05 · metering-interface
existence

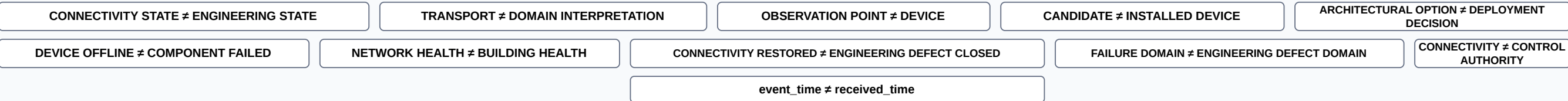
TBD WITH BEELINE — 15
Q-CONN-01 · service models
Q-CONN-02 · SIM/eSIM lifecycle
Q-CONN-03 · exposed states
Q-CONN-04 · delivery semantics
Q-CONN-05 · retry/reconnect ownership
Q-CONN-06 · authentication model
Q-CONN-07 · private connectivity
Q-CONN-08 · RF/site-survey support
Q-CONN-09 · operator observability
Q-CONN-10 · incident exposure
Q-CONN-11 · provisioning interfaces
Q-CONN-12 · lifecycle events
Q-CONN-13 · onboarding role
Q-CONN-14 · edge/gateway ownership
Q-CONN-15 · pilot monitoring

FAILURE / UNKNOWN BOUNDARY · COMPACT STRIP

DEVICE SIDE → LOCAL POWER / OPTIONAL AGGREGATION → ACCESS / RADIO → OPERATOR TRANSPORT → HOUSEMASTER INTEGRATION EDGE → HOUSEMASTER PROCESSING

NO MESSAGE ≠ NORMAL ENGINEERING CONDITION · CONNECTIVITY LOSS ≠ ENGINEERING DEFECT · DELIVERY FAILURE ≠ PHYSICAL FAILURE · UNKNOWN ≠ ABSENT

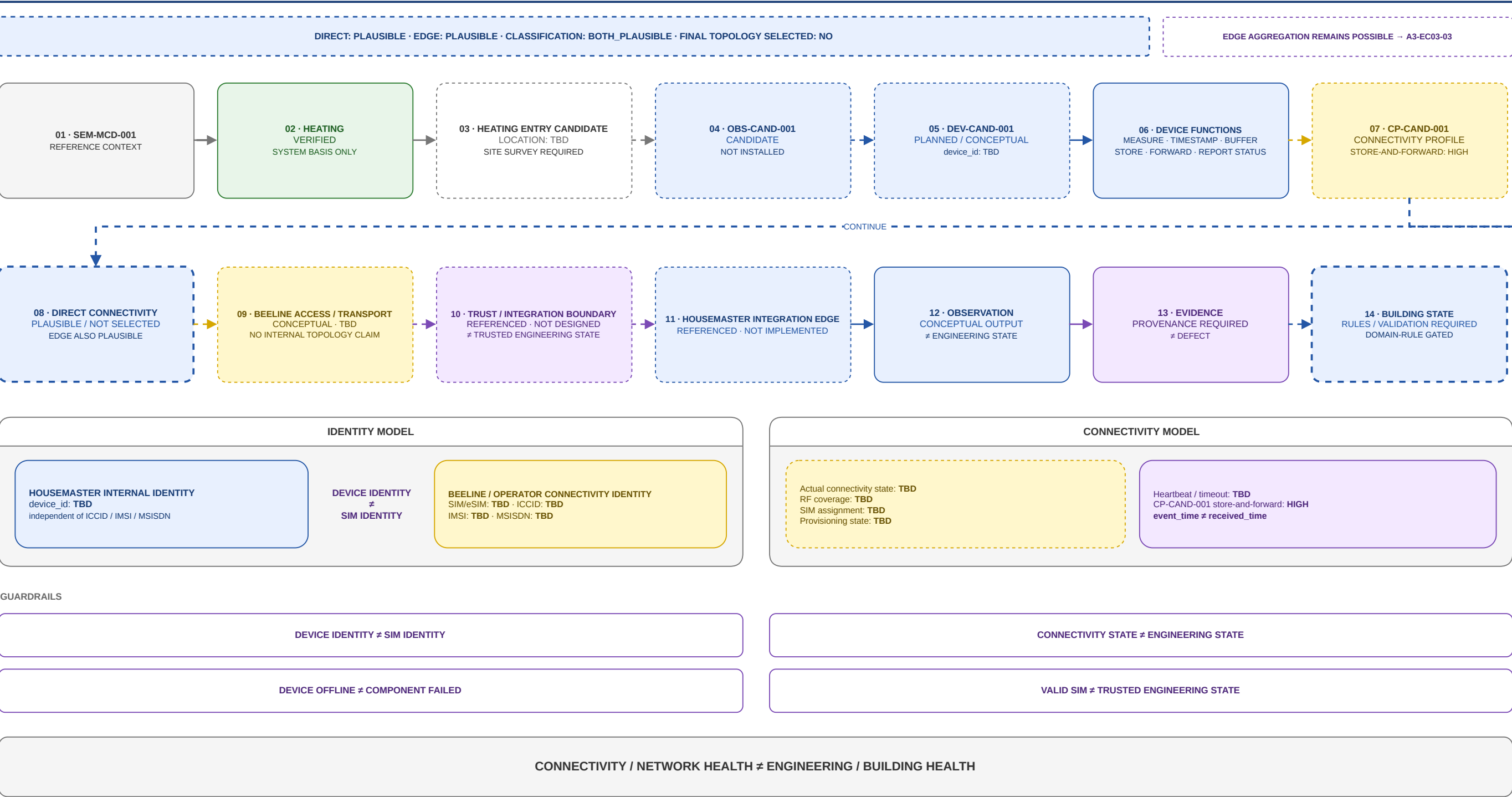
MANDATORY GUARDRAILS · 11



A3-EC03-02 · DIRECT CONNECTIVITY PATTERN

P1-06 · FLOW IR — FROZEN

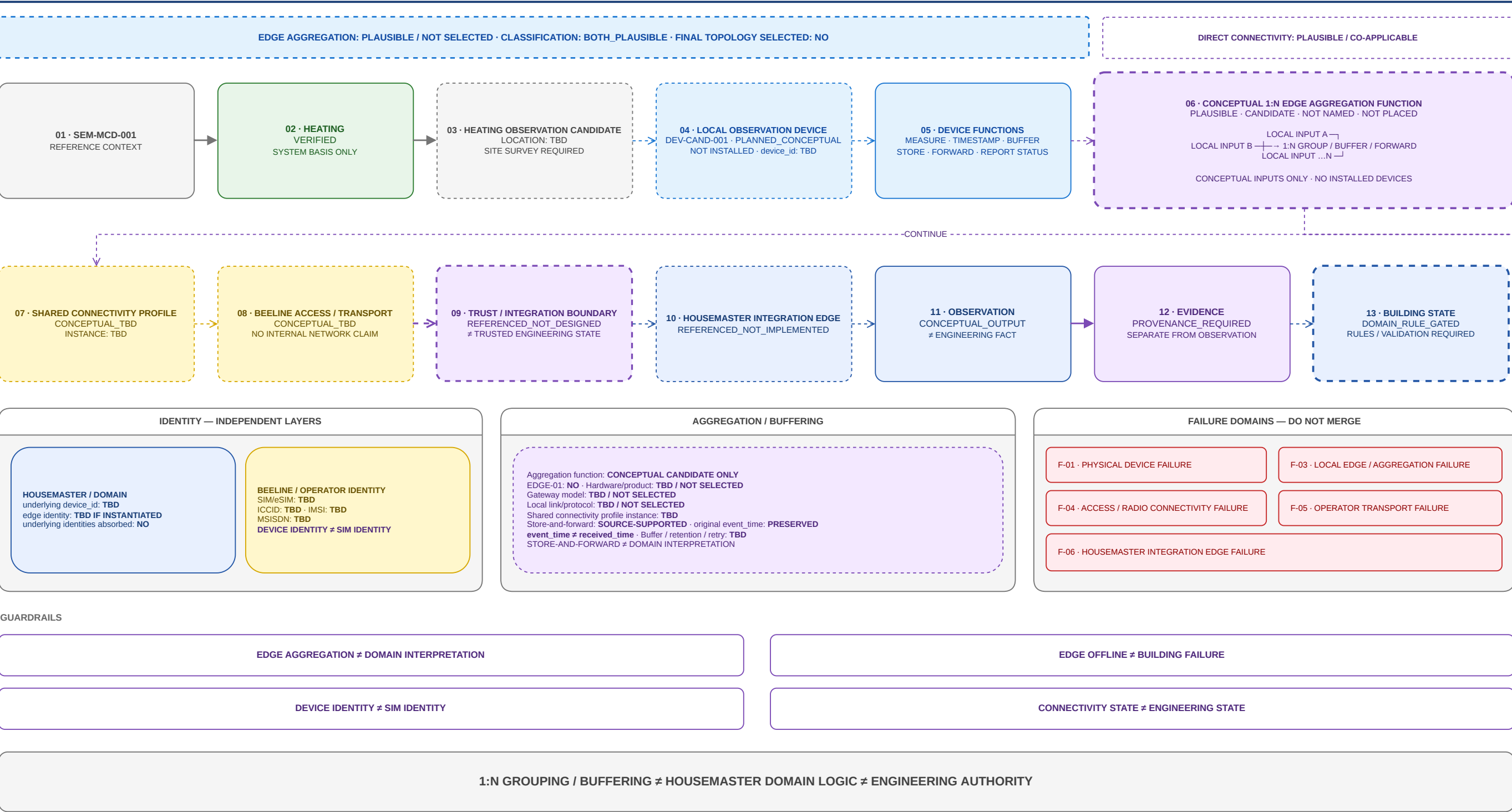
One plausible deployment pattern; identity, connectivity, trust and engineering state remain separate.



A3-EC03-03 · EDGE AGGREGATION PATTERN

P1-07 · FLOW IR — FROZEN

Aggregation groups and forwards observations; it does not absorb device identities, interpret the domain, or establish engineering truth.



FROZEN FLOW · 13 STEPS · 12 EDGES · 4 GUARDRAILS

Direct vs Edge Matrix

Fifteen criteria are compared; no topology winner is declared.

CO-APPLICABLE / NOT SEQUENTIAL / NOT SELECTED

CRITERION		DIRECT CONNECTIVITY PLAUSIBLE ALTERNATIVE	EDGE AGGREGATION PLAUSIBLE ALTERNATIVE
01 Operator-facing connectivity identity count		HIGH — one per device	LOW — one per aggregation group
02 Device independence		HIGH — each device operates without dependency on another local device	LOW — devices depend on the shared aggregation function's availability
03 Local power dependency		LOW — only the device itself needs power	MEDIUM–HIGH — aggregation function also needs power; exact value TBD pending site survey
04 Buffering		Per-device (distributed)	Centralized at aggregation function or still per-device — TBD
05 Aggregation		Not applicable	Native to this pattern
06 Physical distribution		Devices spread across building without a common local node	Devices cluster around a common local node
07 Maintenance complexity		MEDIUM — more individual connectivity endpoints to manage	MEDIUM — fewer connectivity endpoints, one added local component to maintain
08 Lifecycle complexity		MEDIUM — device lifecycle only	MEDIUM–HIGH — device plus aggregation-function lifecycle
09 Failure-domain size		LOW per device — isolated failure	HIGHER for group — aggregation-function failure affects all devices behind it RELATED: A3-EC03-06 / EC-03 §29 F-03
10 Single-point-of-failure exposure		LOW	HIGHER — aggregation function is a candidate single point of failure for its group
11 Local protocol dependency		NONE — no local link required	TBD — unspecified local link required between devices and aggregation function
12 Operator visibility		Per-device connectivity state	Per-aggregation-group connectivity state; device-level detail potentially reduced
13 Site-survey dependency		Per-device location only	Per-device location plus aggregation-function location and local-link feasibility
14 Scalability		MEDIUM — each additional device adds its own connectivity identity	HIGH within cluster — devices can join existing aggregation function without new external connectivity identities
15 Replacement impact		LOW — replacing one device does not affect others	MEDIUM — replacing aggregation function could affect all devices behind it; replacing one device does not

OPEN / TBD — 4 ITEMS

EC03-04-TBD-01 Edge local-power dependency exact value; site survey required.
EC03-04-TBD-02 Edge buffering placement; joint device/integration validation required.
EC03-04-TBD-03 Edge local-link dependency / site feasibility; joint validation required, including Beeline where its boundary is affected.
EC03-04-TBD-04 Direct versus Edge selection = NOT SELECTED; later decision gate required.

RATING VOCABULARY LOW / MEDIUM / HIGH / TBD
Source-preserved qualifiers: MEDIUM–HIGH / HIGHER / NONE / Not applicable. Qualifiers and reasoning are not a calculated score or severity heatmap.

RESPONSIBILITY CONTEXT
Beeline — CONNECTIVITY / TRANSPORT PARTNER
HouseMaster — INDEPENDENT DOMAIN LAYER / SYSTEM OF RECORD
Trust/API Boundary NOT SHOWN. No EDGE-01 instantiation. No deployment inference.

NO TOPOLOGY WINNER DECLARED • NO TOPOLOGY RECOMMENDED OR SELECTED • NO COST FIGURE USED • ARCHITECTURAL OPTION ≠ DEPLOYMENT DECISION

REFERENCE SEM-MCD-001 — Semey, Shakarima 13A | Installed devices: 0 CONFIRMED | Actual installed-device presence: NOT ESTABLISHED | 0 CONFIRMED ≠ PROVED ABSENCE

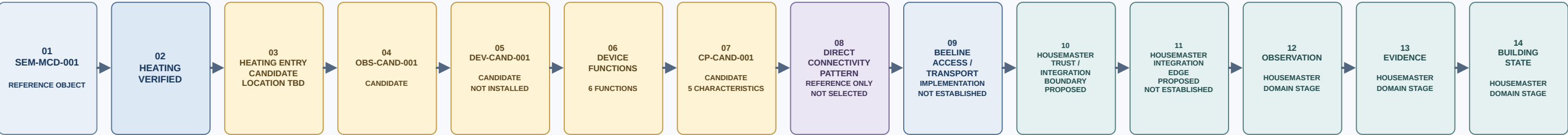
REFERENCE HEATING FLOW

Worked semantic trace — NOT a deployment diagram

P4-04 • ASSEMBLY 021 • PT-06 REFERENCE FLOW • IR FROZEN • PAGE_SPEC_READY

HEATING VERIFIED ≠ HEATING ENTRY LOCATION ESTABLISHED • REFERENCE DIRECT FLOW ≠ DIRECT TOPOLOGY SELECTION

PRIMARY FLOW • 14 STAGES • 13 DIRECTED HANDOFFS • SOURCE ORDER LEFT → RIGHT



ZONE A • REFERENCE ORIGIN

SEM-MCD-001 — Semey, Shakarima 13A
HEATING = VERIFIED
HEATING ENTRY = CANDIDATE / LOCATION TBD

HEATING VERIFIED ≠ HEATING ENTRY LOCATION ESTABLISHED
No installed equipment is shown or inferred.

ZONE B • OBSERVATION / DEVICE / CONNECTIVITY CANDIDATES

OBS-CAND-001 — CANDIDATE | **DEV-CAND-001** — CANDIDATE / NOT INSTALLED DEVICE

DEVICE FUNCTIONS — 6
MEASURE / TIMESTAMP / BUFFER / STORE / FORWARD / REPORT STATUS

CP-CAND-001 — CANDIDATE / 5 CHARACTERISTICS
LATENCY = MEDIUM • LOSS TOLERANCE = MEDIUM • STORE-AND-FORWARD = HIGH
REACHABILITY IMPORTANCE = MEDIUM • MOBILITY = FIXED
Qualitative values are not normalized into numeric scoring.

ZONE C • DIRECT REFERENCE PATTERN

DIRECT CONNECTIVITY PATTERN
PROPOSED ARCHITECTURAL ALTERNATIVE
REFERENCE WALK-THROUGH ONLY
NOT SELECTED

REFERENCE DIRECT FLOW ≠ DIRECT TOPOLOGY SELECTION
Edge is not rejected.

ZONE D • BEELINE HANDOFF

BEELINE ACCESS / TRANSPORT
CONNECTIVITY / TRANSPORT DOMAIN
IMPLEMENTATION NOT ESTABLISHED

BEELINE ACCESS / TRANSPORT ≠ BEELINE INTERNAL IMPLEMENTATION
No internal network detail is shown.

ZONE E • HOUSEMASTER BOUNDARY + DOMAIN

HOUSEMASTER TRUST / INTEGRATION BOUNDARY — PROPOSED BOUNDARY / IMPLEMENTATION NOT ESTABLISHED
HOUSEMASTER INTEGRATION EDGE — PROPOSED ARCHITECTURE / IMPLEMENTATION NOT ESTABLISHED

OBSERVATION → EVIDENCE → BUILDING STATE — three separate semantic stages; no automatic verification, defect confirmation, or connectivity-to-building-state inference.

HOUSEMASTER TRUST / INTEGRATION BOUNDARY ≠ IMPLEMENTED API OR SECURITY DESIGN

OPEN / TBD — 1

EC03-07-TBD-01
Heating entry candidate location = **TBD / NOT ESTABLISHED**
Validation owner = **NOT ESTABLISHED IN §26**

No room, entrance, riser, node, mounting location, or survey result is inferred.

GUARDRAILS — EXACTLY 6

1 HEATING VERIFIED ≠ HEATING ENTRY LOCATION ESTABLISHED

2 OBS-CAND-001 / DEV-CAND-001 / CP-CAND-001 = CANDIDATES, NOT DEPLOYED INSTANCES

3 REFERENCE DIRECT FLOW ≠ DIRECT TOPOLOGY SELECTION

4 BEELINE ACCESS / TRANSPORT ≠ BEELINE INTERNAL IMPLEMENTATION

5 HOUSEMASTER TRUST / INTEGRATION BOUNDARY ≠ IMPLEMENTED API OR SECURITY DESIGN

6 NO REAL NETWORK TECHNOLOGY / PROTOCOL / PRODUCT IS INSTANTIATED

RESPONSIBILITY DOMAINS — 2
BEELINE = CONNECTIVITY / TRANSPORT DOMAIN | **HOUSEMASTER** = TRUST / INTEGRATION BOUNDARY + INTEGRATION EDGE + OBSERVATION + EVIDENCE + BUILDING STATE
PLANE COUNT = 0. DATA / CONTROL / OPERATIONS planes are not rendered.

REFERENCE FLOW — NOT DEPLOYED ARCHITECTURE • No installed equipment, SIM inventory, gateway, APN/private network, protocol, coverage, RF result, API/security implementation, cloud service, 3D, ALAU AI, or commercial content.

REFERENCE SEM-MCD-001 — Semey, Shakarima 13A | HEATING: VERIFIED | HEATING ENTRY LOCATION: TBD / NOT ESTABLISHED | SOURCE: EC-03 §26

A3-EC03-06 · FAILURE DOMAINS

Seven distinct failure domains — each defines an observable effect, operational boundary and explicit non-proof.

F-01 ≠ F-02 ≠ F-03 ≠ F-04 ≠ F-05 ≠ F-06 ≠ F-07 · ALL FAILURE DOMAINS ≠ ENGINEERING COMPONENT FAILURE ≠ BUILDING FAILURE						
FAILURE DOMAIN	RESPONSIBILITY LAYER	OBSERVABLE EFFECT	DATA / CONNECTIVITY CONSEQUENCE	WHAT REMAINS OPERATIONAL	WHAT IT DOES NOT PROVE	RECOVERY / NEXT CHECK
F-01 PHYSICAL DEVICE FAILURE	BUILDING / FIELD	Device silent or reports an implausible value.	That device's observations unavailable or potentially invalid; evidence gap.	Last known building state; observations from other devices.	Engineering component failure or building failure.	Root cause / recovery: unresolved. Buffering alone does not prove device recovery.
F-02 LOCAL POWER FAILURE	BUILDING / FIELD	Multiple devices unreachable together; power correlation if present.	Affected devices' observations unavailable; circuit-wide evidence gap.	Devices on unaffected circuits; last known building state.	Failure of every affected component or the building.	Power restoration / restart / retained buffer behavior: unresolved.
F-03 LOCAL EDGE / AGGREGATION FAILURE	BUILDING / FIELD or JOINT if operator-managed	Multiple devices unreachable together, correlated by physical cluster.	All observations behind that aggregation function unavailable; cluster gap.	Direct-connected devices; other aggregation groups; last known state.	Underlying device, component or building failure.	Recovery not designed. Buffered delivery, if any, preserves original event_time.
F-04 ACCESS / RADIO CONNECTIVITY FAILURE	TBD WITH BEELINE	HouseMaster integration state becomes UNREACHABLE or DEGRADED.	Observations temporarily unavailable or delayed; temporary evidence gap.	Last known building state; local observation/buffering where supported.	Root cause, device/component failure or building failure.	Forward buffered observations after recovery; preserve event_time. Root cause / values: TBD.
F-05 OPERATOR TRANSPORT FAILURE	BEELINE	Correlated connectivity loss beyond one building's pattern.	Observations across shared paths unavailable/delayed; broad evidence gap/backlog.	Last known building state; local observation/buffering where supported.	Device/component/building failure or engineering state change.	Forward buffered observations after recovery; preserve event_time. Retry/SLA: undefined.
F-06 HOUSEMASTER INTEGRATION EDGE FAILURE	HOUSEMASTER	Ingress/processing health signals; connectivity may remain healthy.	New observations cannot be ingested; acceptance delayed; ingestion-side gap.	Already-ingested building state; upstream device/connectivity functions.	Beeline transport, device, component or building failure.	Recovery not designed. Delayed buffered delivery preserves original event_time.
F-07 HOUSEMASTER PROCESSING FAILURE	HOUSEMASTER	Processing backlog or latency.	Domain interpretation and workflow progression may be delayed.	Raw ingested observations if durably stored; existing building state.	Device/connectivity, engineering component or building failure.	Processing recovery undefined. Raw data remains only if durably stored.

HOUSEMASTER INTEGRATION STATES — NOT BEELINE INTERNAL NETWORK STATES

UNKNOWN · PROVISIONED · REACHABLE · UNREACHABLE · DEGRADED · SUSPENDED · DECOMMISSIONED

UNREACHABLE = EXPECTED PATH NOT RESPONDING · ROOT CAUSE UNCLASSIFIED

UNREACHABLE is not mapped to one failure domain and does not identify device, power, edge, access, transport or integration root cause.

TIMING / STORE-AND-FORWARD — SOURCE-SUPPORTED

event_time CREATED → LOCAL BUFFER / STORE → TRANSPORT UNAVAILABLE → CONNECTIVITY RESTORED → BUFFERED OBSERVATION FORWARDED → received_time CREATED

event_time ≠ received_time WHEN BUFFERED · DELAYED DELIVERY DOES NOT REWRITE event_time

CONNECTIVITY LOSS MAY CREATE TEMPORARY EVIDENCE GAP · CONNECTIVITY LOSS ≠ ENGINEERING FAILURE

Buffer size: NOT DEFINED · Retention: NOT DEFINED · Retry: NOT DEFINED · Heartbeat / timeout / polling: NOT DEFINED

DEVICE OFFLINE ≠ COMPONENT FAILED

EDGE OFFLINE ≠ BUILDING FAILURE

NETWORK HEALTH ≠ BUILDING HEALTH

CONNECTIVITY STATE ≠ ENGINEERING STATE

CONNECTIVITY STATE / REACHABILITY

Connectivity state tells HouseMaster what is known about the path; it does not by itself establish physical or engineering condition.

PROPOSED HOUSEMASTER INTEGRATION STATES ≠ BEELINE INTERNAL NETWORK STATES

ZONE B • HOUSEMASTER CONNECTIVITY-STATE VOCABULARY • 7 SEMANTIC PEERS • NO ORDER / NO TRANSITIONS / NO SEVERITY SCALE

UNKNOWN No connectivity information has ever been received for this device/profile.	PROVISIONED A connectivity profile is conceptually associated with a device; reachability not yet confirmed.	REACHABLE Connectivity path currently confirmed responsive.	UNREACHABLE Connectivity path currently not responding; cause not yet classified.	DEGRADED Connectivity path responsive but reduced quality, such as delayed delivery, without being fully unreachable.	SUSPENDED Connectivity intentionally paused by administrative action. NOT A FAULT.	DECOMMISSIONED Device/connectivity profile retired and no longer expected to report.
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ZONE C • REACHABILITY / KNOWLEDGE MATRIX • TRANSPORT EVIDENCE ≠ ENGINEERING-CONDITION DETERMINATION

CONDITION	OBSERVATION STATE	HOUSEMASTER MAY KNOW	HOUSEMASTER MUST NOT INFER
01 NO ENGINEERING OBSERVATION RECEIVED	No new data since last known state	No update has arrived	Underlying engineering condition is unchanged, healthy or failed
02 DEVICE UNREACHABLE	Connectivity path not responding	Connectivity path is down	Physical device or engineering component has failed
03 DEVICE REACHABLE BUT NO ENGINEERING EVENT	Connectivity confirmed responsive, but no telemetry/event content	Transport path itself is functioning	Absence of an event necessarily means normal. Event-driven: silence may be expected. Continuous/periodic: silence may indicate an unclassified device/interface issue.
04 DEVICE REACHABLE AND OBSERVATION DELIVERED	Connectivity and content both confirmed	Reported value/event is available for domain interpretation	Delivered value is automatically a verified engineering fact. Preserve: Observation → Symptom → Defect gating.

ZONE D • RESPONSIBILITY BOUNDARY

HOUSEMASTER — integration-state vocabulary; permitted-knowledge semantics; domain interpretation; evidence gating.
BEELINE — connectivity / transport domain.

HOUSEMASTER CONNECTIVITY STATES ≠ BEELINE INTERNAL NETWORK STATES
No Beeline state mapping is inferred.

ZONE E • OPEN / TBD — 4

EC03-05-TBD-01 Joint validation of seven-state HouseMaster vocabulary
EC03-05-TBD-02 Heartbeat interval = NOT DEFINED
EC03-05-TBD-03 Timeout value = NOT DEFINED
EC03-05-TBD-04 Polling frequency = NOT DEFINED
No value is inferred or supplied.

ZONE F • GUARDRAILS — 6

HOUSEMASTER CONNECTIVITY STATES ≠ BEELINE INTERNAL NETWORK STATES	NO UPDATE ≠ UNCHANGED / HEALTHY / FAILED ENGINEERING CONDITION	UNREACHABLE CONNECTIVITY PATH ≠ FAILED PHYSICAL DEVICE OR ENGINEERING COMPONENT
REACHABLE WITH NO EVENT ≠ NECESSARILY NORMAL	DELIVERED OBSERVATION ≠ VERIFIED ENGINEERING FACT	NO HEARTBEAT INTERVAL / TIMEOUT / POLLING FREQUENCY IS DEFINED

PROVENANCE ANCHOR ONLY • SEM-MCD-001 — Semey, Shakarima 13A • No installed-device, SIM, gateway, availability, heartbeat-behavior, or site-condition inference

MASTER DEVICE & CONNECTIVITY MATRIX

Provider-neutral engineering matrix — NOT deployed inventory / topology selection / provider architecture / installation plan

16 CANDIDATE RECORDS × 12 ATTRIBUTE DIMENSIONS = 192 RELATIONSHIPS • PHYSICAL READY 0 • CONCEPTUAL READY 16 • FINAL TOPOLOGY SELECTED 0

BAND A — IDENTITY / OBSERVABLE / FUNCTION / TRAFFIC • ROWS 001–016						
DEV ID	OBS ID	OBSERVABLE CONDITION	FUNCTION SET	DEVICE CLASS	TRAFFIC PATTERN	UPLINK / DOWNLINK
DEV-CAND-001	OBS-CAND-001	Temperature condition (heating entry)	FSET-MEASURE	D-CLASS-01	PERIODIC TELEMETRY	UPLINK ONLY
DEV-CAND-002	OBS-CAND-002	Pressure condition (heating entry)	FSET-MEASURE	D-CLASS-01	PERIODIC TELEMETRY	UPLINK ONLY
DEV-CAND-003	OBS-CAND-003	Flow/consumption condition (heating metering)	FSET-READ	D-CLASS-03	COUNTER/METERING	UPLINK + DOWNLINK
DEV-CAND-004	OBS-CAND-004	Pressure condition (CW entry)	FSET-MEASURE	D-CLASS-01	PERIODIC TELEMETRY	UPLINK ONLY
DEV-CAND-005	OBS-CAND-005	Flow condition (CW entry)	FSET-MEASURE	D-CLASS-01	PERIODIC TELEMETRY	UPLINK ONLY
DEV-CAND-006	OBS-CAND-006	Flow/consumption condition (CW metering)	FSET-READ	D-CLASS-03	COUNTER/METERING	UPLINK + DOWNLINK
DEV-CAND-007	OBS-CAND-007	Temperature/availability condition (hot water)	FSET-MEASURE	D-CLASS-01	PERIODIC TELEMETRY	UPLINK ONLY
DEV-CAND-008	OBS-CAND-008	Leakage indication (hot water)	FSET-DETECT	D-CLASS-04	STATE CHANGE	UPLINK ONLY
DEV-CAND-009	OBS-CAND-009	Water-presence/flooding indication (basement)	FSET-DETECT	D-CLASS-04	EVENT-DRIVEN	UPLINK ONLY
DEV-CAND-010	OBS-CAND-010	Temperature condition (basement)	FSET-MEASURE	D-CLASS-02	PERIODIC TELEMETRY	UPLINK ONLY
DEV-CAND-011	OBS-CAND-011	Humidity condition (basement)	FSET-MEASURE	D-CLASS-02	PERIODIC TELEMETRY	UPLINK ONLY
DEV-CAND-012	OBS-CAND-012	Supply-availability condition (electrical)	FSET-DETECT	D-CLASS-04	STATE CHANGE	UPLINK ONLY
DEV-CAND-013	OBS-CAND-013	Outage/event condition (electrical)	FSET-DETECT	D-CLASS-04	EVENT-DRIVEN	UPLINK ONLY
DEV-CAND-014	OBS-CAND-015	Moisture/water-ingress condition (roof)	FSET-DETECT	D-CLASS-04	EVENT-DRIVEN	UPLINK ONLY
DEV-CAND-015	OBS-CAND-017	Availability/operational condition (lift)	FSET-DETECT	D-CLASS-04	STATE CHANGE	UPLINK ONLY
DEV-CAND-016	OBS-CAND-018	Outage/event condition (lift)	FSET-DETECT	D-CLASS-04	EVENT-DRIVEN	UPLINK ONLY

BAND B — CONNECTIVITY / PATTERN / READINESS • SYNCHRONIZED DEV KEY						
DEV ID	PROFILE	PATTERN CLASS	STORE / FWD	REACHABILITY	PHYSICAL READINESS	CONCEPTUAL READINESS
DEV-CAND-001	CP-CAND-001	BOTH PLAUSIBLE	HIGH	MEDIUM	L1 — NOT READY	READY
DEV-CAND-002	CP-CAND-001	BOTH PLAUSIBLE	HIGH	MEDIUM	L1 — NOT READY	READY
DEV-CAND-003	CP-CAND-002	BOTH PLAUSIBLE	HIGH	LOW	L1 — NOT READY	READY
DEV-CAND-004	CP-CAND-001	BOTH PLAUSIBLE	HIGH	MEDIUM	L1 — NOT READY	READY
DEV-CAND-005	CP-CAND-001	BOTH PLAUSIBLE	HIGH	MEDIUM	L1 — NOT READY	READY
DEV-CAND-006	CP-CAND-002	BOTH PLAUSIBLE	HIGH	LOW	L1 — NOT READY	READY
DEV-CAND-007	CP-CAND-001	DIRECT-FRIENDLY	HIGH	MEDIUM	L1 — NOT READY	READY
DEV-CAND-008	CP-CAND-003	DIRECT-FRIENDLY	HIGH	HIGH	L1 — NOT READY	READY
DEV-CAND-009	CP-CAND-003	EDGE-FRIENDLY	HIGH	HIGH	L1 — NOT READY ZONE-EXISTENCE-GATED	READY
DEV-CAND-010	CP-CAND-004	EDGE-FRIENDLY	MEDIUM	LOW	L1 — NOT READY ZONE-EXISTENCE-GATED	READY
DEV-CAND-011	CP-CAND-004	EDGE-FRIENDLY	MEDIUM	LOW	L1 — NOT READY ZONE-EXISTENCE-GATED	READY
DEV-CAND-012	CP-CAND-003	BOTH PLAUSIBLE	HIGH	HIGH	L1 — NOT READY	READY
DEV-CAND-013	CP-CAND-003	BOTH PLAUSIBLE	HIGH	HIGH	L1 — NOT READY	READY
DEV-CAND-014	CP-CAND-001	TBD	HIGH	MEDIUM	L1 — NOT READY ACCESS-GATED	READY
DEV-CAND-015	CP-CAND-003	EDGE-FRIENDLY	HIGH	HIGH	L1 — NOT READY	READY
DEV-CAND-016	CP-CAND-003	EDGE-FRIENDLY	HIGH	HIGH	L1 — NOT READY	READY

FUNCTION SET LEGEND — EXACT
FSET-MEASURE = MEASURE / TIMESTAMP / BUFFER / STORE / FORWARD / REPORT STATUS
FSET-READ = READ INTERFACE / COUNT / TIMESTAMP / BUFFER / STORE / FORWARD / REPORT STATUS
FSET-DETECT = DETECT / TIMESTAMP / BUFFER / STORE / FORWARD / REPORT STATUS
UPLINK ONLY = 14 • UPLINK + DOWNLINK = 2 • UPLINK + DOWNLINK ≠ ENGINEERING ACTUATION

SOURCE-DERIVED SUMMARY
CANDIDATES 16 • INSTALLED DEVICES ESTABLISHED 0
PHYSICAL READY 0 • L1 — NOT READY 16 • CONCEPTUAL READY 16
DIRECT-FRIENDLY 2 • EDGE-FRIENDLY 5 • BOTH PLAUSIBLE 8 • TBD 1
FINAL TOPOLOGY SELECTED 0
READY ≠ IMPLEMENTED OR DEPLOYED
L1 — NOT READY ≠ PROVED ABSENCE

EC03-08-TBD-01
Physical readiness / candidate location across all 16 = **L1 — NOT READY / NOT ESTABLISHED**
Includes DEV-009 / 010 / 011 zone-existence gates and DEV-014 roof access gate.
Owner: NOT ESTABLISHED IN ASSIGNED SOURCE SCOPE

EC03-08-TBD-02
Final Direct/Edge topology = **NOT SELECTED** • DEV-CAND-014 pattern class = **TBD**
Owner: NOT ESTABLISHED IN ASSIGNED SOURCE SCOPE

1 DEVICE CANDIDATE ≠ INSTALLED DEVICE

2 L1 — NOT READY ≠ PROVED ABSENCE

3 CONCEPTUAL ARCHITECTURE READY ≠ IMPLEMENTED OR DEPLOYED

4 PATTERN LABELS = PLAUSIBILITY, NOT TOPOLOGY SELECTION

5 CONNECTIVITY PROFILE ≠ SIM / TARIFF / APN / VPN / NETWORK TECHNOLOGY / OPERATOR PRODUCT

6 UPLINK + DOWNLINK ≠ ENGINEERING ACTUATION

7 D-CLASS-05 IS CONCEPTUAL ONLY; NO EDGE-01 IS NAMED OR PLACED

8 MASTER MATRIX VALUES ARE SOURCE-REUSED; NO CELL MAY BE INVENTED OR NORMALIZED

PROVIDER-NEUTRAL • RESPONSIBILITY DOMAINS 0 • PLANES 0 • TRUST / ACCESS / INTEGRATION BOUNDARY NOT APPLICABLE • DIRECT-FRIENDLY ≠ DIRECT SELECTED • EDGE-FRIENDLY ≠ EDGE SELECTED • BOTH PLAUSIBLE = CO-APPLICABLE / NOT SEQUENTIAL / NOT SELECTED

No responsibility swimlanes, Trust boundary, DATA / CONTROL / OPERATIONS planes, winner colors, severity heatmap, deployment icons, Direct/Edge flow, network topology, SIM/device/gateway inventory, ALAU AI, or 3D.

REFERENCE CONTEXT ONLY • SEM-MCD-001 — Semey, Shakarima 13A • No new building fact inferred • SOURCE: EC-03 §33 • DIRECT DEFINITIONS: §§16–19 / §25

SCOPE & OPEN QUESTIONS

Which operator-specific technical assumptions, interfaces, constraints, responsibilities, and pilot parameters must be jointly validated with Beeline before the HouseMaster reference architecture can become a validated pilot architecture?

15 UNRESOLVED QUESTIONS • 7 VALIDATION TRACKS | NONE ANSWERED | OWNER ASSIGNMENTS 0 | NO BEELINE COMMITMENT

TBD WITH BEELINE = Operator-specific implementation detail requiring validation with Beeline.
DOES NOT MEAN: architecture error / missing HouseMaster design / assumed Beeline capability / Beeline commitment.

BEELINE FULL INTERNAL NETWORK / PLATFORM DISCLOSURE IS NOT REQUIRED
Only information needed to validate the agreed integration surface is requested. No Beeline internal architecture is depicted.

TECH-01 CONNECTIVITY / SIGNAL PATH	TECH-03 API / DATA	TECH-05 DEPLOYMENT / CLOUD
Q1 What IoT/M2M connectivity models can Beeline support for a Connected Building pilot? STATUS: TBD WITH BEELINE • DECISION: TBD • OWNER / NOTE: UNASSIGNED / BLANK	Q6 Which operator APIs, events or integration interfaces are available for connectivity management and telemetry/event exchange? STATUS: TBD WITH BEELINE • DECISION: TBD • OWNER / NOTE: UNASSIGNED / BLANK	Q11 What infrastructure or cloud role, if any, should Beeline play in the pilot architecture? STATUS: TBD WITH BEELINE • DECISION: TBD • OWNER / NOTE: UNASSIGNED / BLANK
Q2 Which transport options and connectivity-state events can be exposed to an external domain platform such as HouseMaster? STATUS: TBD WITH BEELINE • DECISION: TBD • OWNER / NOTE: UNASSIGNED / BLANK	Q7 Which data objects may cross the Beeline ↔ HouseMaster boundary, and what restrictions apply? STATUS: TBD WITH BEELINE • DECISION: TBD • OWNER / NOTE: UNASSIGNED / BLANK	Q12 What Kazakhstan data-residency, resilience, backup, DR and deployment constraints should be incorporated into the validated architecture? STATUS: TBD WITH BEELINE • DECISION: TBD • OWNER / NOTE: UNASSIGNED / BLANK
Q3 What network-side constraints should be assumed for basement, technical-room and other difficult indoor MCD environments? STATUS: TBD WITH BEELINE • DECISION: TBD • OWNER / NOTE: UNASSIGNED / BLANK	Q8 Which interaction model is preferred: synchronous API, asynchronous events, webhook/callback, message transport, or another operator-approved mechanism? STATUS: TBD WITH BEELINE • DECISION: TBD • OWNER / NOTE: UNASSIGNED / BLANK	TECH-06 OPERATIONS / SLA
TECH-02 IDENTITY / LIFECYCLE	TECH-04 SECURITY / TRUST	Q13 How should operational responsibility be divided between Beeline connectivity operations and HouseMaster building-domain operations? STATUS: TBD WITH BEELINE • DECISION: TBD • OWNER / NOTE: UNASSIGNED / BLANK
Q4 Which SIM/eSIM/M2M identity objects and lifecycle states can Beeline expose to HouseMaster? STATUS: TBD WITH BEELINE • DECISION: TBD • OWNER / NOTE: UNASSIGNED / BLANK	Q9 What authentication, authorization and trust model would Beeline require for HouseMaster integration? STATUS: TBD WITH BEELINE • DECISION: TBD • OWNER / NOTE: UNASSIGNED / BLANK	Q14 What incident, escalation, observability, support and SLA/SLO interfaces can be agreed for the pilot? STATUS: TBD WITH BEELINE • DECISION: TBD • OWNER / NOTE: UNASSIGNED / BLANK
Q5 What provisioning, activation, suspension, replacement and deactivation interfaces are available or feasible? STATUS: TBD WITH BEELINE • DECISION: TBD • OWNER / NOTE: UNASSIGNED / BLANK	Q10 What security, audit, logging, credential and incident-response requirements must the pilot satisfy? STATUS: TBD WITH BEELINE • DECISION: TBD • OWNER / NOTE: UNASSIGNED / BLANK	TECH-07 PILOT
		Q15 What minimum pilot scope, technical acceptance criteria, participants and duration would Beeline consider sufficient to make a scale-up decision? STATUS: TBD WITH BEELINE • DECISION: TBD • OWNER / NOTE: UNASSIGNED / BLANK

ARCHITECTURE-VALIDATION OUTCOMES — 4 EQUAL OPTIONS

CONFIRM — Reference model applies | REPLACE — Use Beeline actual / alternative mechanism
CONSTRAIN — Apply technical / security / operational limit | REMOVE — Not used in pilot
Not commercial / approval / contract / deployment statuses. No ranking, score, winner, or traffic-light semantics.

VALIDATION PIPELINES

HOUSEMASTER REFERENCE ARCHITECTURE + BEELINE TECHNICAL INPUT → JOINT VALIDATION → VALIDATED PILOT ARCHITECTURE

TBD → CONFIRM / REPLACE / CONSTRAIN / REMOVE → PILOT DESIGN
VALIDATED PILOT ARCHITECTURE ≠ PRODUCTION ARCHITECTURE

EXPECTED AFTER JOINT VALIDATION — 7

1 Confirmed integration capabilities
2 Agreed alternative mechanisms
3 Documented technical constraints
4 Removed out-of-scope assumptions
5 Assigned unresolved TBDs
6 Validated integration surface
7 Inputs for Pilot Architecture v0.1

OPEN / TBD STATUS • OPEN/TBD 15 • TBD WITH BEELINE 15 • DECISION = TBD 15 • OWNER ASSIGNMENTS 0 • UNASSIGNED / BLANK IS AN OPEN FIELD, NOT APPROVAL

1 REFERENCE ARCHITECTURE ≠ BEELINE INTERNAL ARCHITECTURE	2 VALIDATED PILOT ARCHITECTURE ≠ PRODUCTION ARCHITECTURE	3 TBD WITH BEELINE ≠ ARCHITECTURE ERROR	4 TBD WITH BEELINE ≠ MISSING HOUSEMASTER DESIGN
5 TBD WITH BEELINE ≠ ASSUMED BEELINE CAPABILITY	6 TBD WITH BEELINE ≠ BEELINE COMMITMENT	7 VALIDATION INPUT / OUTCOME ≠ COMMERCIAL COMMITMENT OR STATUS	

NOT an approval sheet / readiness scorecard / commercial status page / Beeline commitment page / production architecture page. No SEM-MCD-001, scoring, maturity or readiness percentage, severity colors, approval badge, Beeline architecture diagram, deployment diagram, implementation drawing, commercial layer, ALAU AI, or 3D.

TBD WITH BEELINE

A3-EC03-09 · CONNECTIVITY-SPECIFIC VALIDATION-QUESTION REGISTER

PRODUCTION P5-02 · ASSEMBLY 024

TYPE PT-08 TBD / VALIDATION REGISTER

SOURCE EC-03 §30

PRIMARY ENGINEERING QUESTION

Which Beeline-specific connectivity capabilities, externally exposed states, interfaces, constraints, responsibilities, and pilot operations remain unknown and must be validated before EC-03 assumptions can be carried forward?

CONTROLLED OPEN STATE

15 Q-CONN QUESTIONS REMAIN UNANSWERED / UNOWNED / UNDECIDED.

§30 records what each unknown affects. It resolves none of them.

Q ID	FULL QUESTION	WHY NEEDED	STATUS	ANSWER	OWNER	DECISION
Q-CONN-01	What machine-connectivity service models are available for pilot use?	Determines feasible connectivity profiles (§17)	TBD WITH BEELINE	UNANSWERED	UNASSIGNED	NOT RECORDED
Q-CONN-02	What SIM/eSIM lifecycle capabilities are exposed?	Needed for EC-04's identity/lifecycle design	TBD WITH BEELINE	UNANSWERED	UNASSIGNED	NOT RECORDED
Q-CONN-03	What connectivity/reachability states can be exposed externally?	Needed to map §21's proposed states to actual Beeline-exposed states	TBD WITH BEELINE	UNANSWERED	UNASSIGNED	NOT RECORDED
Q-CONN-04	What delivery semantics are available?	Affects §23's store-and-forward assumptions	TBD WITH BEELINE	UNANSWERED	UNASSIGNED	NOT RECORDED
Q-CONN-05	What retry/reconnect behavior is operator-controlled vs. device-controlled?	Affects §23/§29 failure-recovery boundaries	TBD WITH BEELINE	UNANSWERED	UNASSIGNED	NOT RECORDED
Q-CONN-06	What authentication/security models are supported?	TECH-04 territory — referenced only, not designed here	TBD WITH BEELINE	UNANSWERED	UNASSIGNED	NOT RECORDED
Q-CONN-07	What private connectivity options exist if later required?	Affects future connectivity-profile options	TBD WITH BEELINE	UNANSWERED	UNASSIGNED	NOT RECORDED
Q-CONN-08	What basement/technical-room RF/site-survey support is available?	Directly affects basement-cluster candidates DEV-CAND-009 / 010 / 011	TBD WITH BEELINE	UNANSWERED	UNASSIGNED	NOT RECORDED
Q-CONN-09	What operator-side observability can HouseMaster receive?	Affects §21's state-mapping accuracy	TBD WITH BEELINE	UNANSWERED	UNASSIGNED	NOT RECORDED
Q-CONN-10	How are connectivity incidents exposed?	Affects future TECH-06-aligned operations design	TBD WITH BEELINE	UNANSWERED	UNASSIGNED	NOT RECORDED
Q-CONN-11	What service-provisioning interfaces exist?	Needed for EC-04's provisioning lifecycle	TBD WITH BEELINE	UNANSWERED	UNASSIGNED	NOT RECORDED
Q-CONN-12	What lifecycle events are available for SIM/eSIM?	Needed for EC-04's SIM/eSIM lifecycle inventory (§31)	TBD WITH BEELINE	UNANSWERED	UNASSIGNED	NOT RECORDED
Q-CONN-13	What role can Beeline play in device/gateway onboarding?	Affects Edge Aggregation feasibility (§25/§28)	TBD WITH BEELINE	UNANSWERED	UNASSIGNED	NOT RECORDED
Q-CONN-14	What edge/gateway ownership patterns are acceptable?	Affects F-03's responsibility assignment (§29)	TBD WITH BEELINE	UNANSWERED	UNASSIGNED	NOT RECORDED
Q-CONN-15	What pilot monitoring/escalation model is feasible?	TECH-06/TECH-07 territory — referenced only	TBD WITH BEELINE	UNANSWERED	UNASSIGNED	NOT RECORDED

OPEN / TBD CONTROL TOTALS

1500015
UNANSWEREDANSWEREDOWNERSDECISIONSTRACKS / CATEGORIESOPEN / TBD

DEPENDENCY DISCIPLINE

Each question has exactly one source-explicit **WHY NEEDED** dependency. References to TECH-04 / TECH-06 / TECH-07 / EC-04 / §§17, 21, 23, 25, 28, 29, 31 are dependency references only — not semantic groups.
QUESTION → DEPENDENCY RELATIONSHIPS: 15

SOURCE GUARDRAIL — 1

EVERY QUESTION IS UNANSWERED BY DESIGN — THIS DOCUMENT DOES NOT RESOLVE ANY OF THEM.

PROHIBITED INFERENCE

No answer · owner · decision · agreement · approval · Beeline commitment · topology selection · provider architecture · deployment readiness. No invented SIM/eSIM capability, API, security, private connectivity, cloud, SLA/SLO, pilot parameter, or Beeline internal architecture.

A3-EC01-05 CONFLICTS / TBDs / CONFIDENCE

REFERENCE SEM-MCD-001 — Semey, Shakarima 13A

AN OPEN ITEM ≠ A DATA ERROR
Unresolved information is a controlled engineering state until source-backed evidence or field validation closes it.

PRIMARY VISUAL — CONFLICT / TBD / CONFIDENCE REGISTER • 12 CONTROLLED OPEN ITEMS

ITEM	CURRENT STATUS	CONFIDENCE	WHY OPEN / SOURCE	REQUIRED CLOSURE	DOWNSTREAM RESTRICTION
CI-01 EXACT BUILDING FOOTPRINT	NOT ESTABLISHED / TBD	TBD	No dimensioned geometry or field survey. EC-01 §4 / BF-008 / TBD-001	AUTHORITATIVE GEOMETRY / FIELD SURVEY	DO NOT TREAT 3D AS VERIFIED AS-BUILT GEOMETRY
CI-02 STEPPED / NON-RECTANGULAR GEOMETRY	PARTIAL	PARTIAL	Shape evidenced; full dimensions incomplete. SRC-004 Image 3 / BF-009	AUTHORITATIVE GEOMETRY / MEASUREMENT	DO NOT FINALIZE FOOTPRINT / MASSING DIMENSIONS
CI-03 HEAT-03	NO EVIDENCE	NO EVIDENCE	System exists; component evidence absent. EC-01 §13	SOURCE-BACKED EVIDENCE OR EXPLICIT REJECTION	DO NOT INFER EXISTENCE OR ABSENCE
CI-04 APARTMENT COUNT — 81 VS 78	DISPUTED / OPEN	DISPUTED / OPEN	Compiled 81; alternate 78 not resolved. BF-014 / CONFLICT-005	AUTHORITATIVE RECORD / RESOLVE SOURCE CONFLICT	DO NOT SILENTLY CHOOSE 78 OR 81
CI-05 LAND-LETTER MISMATCH	OPEN	DISPUTED / OPEN	Secondary citation differs from official letter. CONFLICT-004	AUTHORITATIVE CADASTRAL / LAND DOCUMENTATION	DO NOT TREAT SECONDARY LEGAL CITATION AS CONFIRMED
CI-06 BASEMENT / TECHNICAL-SPACE PLAN + HEIGHT	TBD	TBD	No plan, exact height, or completed survey. TBD-003	SITE SURVEY / AUTHORITATIVE PLAN	DO NOT MODEL VERIFIED BASEMENT GEOMETRY OR SELECT INSTALLATION POINT
CI-07 ENGINEERING NODE / COMPONENT LOCATIONS	TBD	TBD	No confirmed node/component locations. EC-01 §11 / TBD-005	SITE SURVEY / ENGINEERING DOCUMENTATION	DO NOT ASSIGN INSTALLATION LOCATION
CI-08 OBSERVATION / DEVICE LOCATIONS + CONSTRAINTS	TBD	TBD	Mounting, power, environment and access unknown. EC-01 §11	SITE SURVEY	DO NOT CLAIM INSTALLED DEVICE OR CONFIRMED MOUNTING LOCATION
CI-09 CONNECTIVITY FEASIBILITY AT CANDIDATE POINTS	TBD	TBD	No field/network test at candidate points. EC-01 §10–§11	FIELD / NETWORK VALIDATION WITH BEELINE	DO NOT SELECT DIRECT OR EDGE TOPOLOGY
CI-10 FACADE / ELEVATIONS	TBD	TBD	No authoritative elevations / complete capture. TBD-004	AUTHORITATIVE ELEVATIONS / FIELD CAPTURE	DO NOT FINALIZE FACADE PROPORTIONS / GEOMETRY
CI-11 INVENTORY NUMBER / YEAR CONFIRMATION	PARTIAL / TBD	PARTIAL	23161 / 1978 lack primary confirmation. BF-004 / BF-005 / TBD-008	CONFIRM SOURCE IDENTITY RECORDS	DO NOT PRESENT BOTH IDENTITY FIELDS AS FULLY VERIFIED
CI-12 SOURCE-REGISTER CONTENT	TBD	TBD	Artifact content unreadable / unvalidated. SRC-008 / TBD-009	VALIDATE SOURCE-REGISTER ARTIFACT	DO NOT USE REGISTER TO SUPERSEDE OR CORROBORATE FACTS

ACTIVE CONFLICTS — 2 (NOT RESOLVED)

CONFLICT-004
LAND-LETTER REFERENCE NUMBER / DATE — OPEN

CONFLICT-005
APARTMENT COUNT 81 VS 78 — OPEN

CONFIDENCE / EVIDENCE STATES — 7 UNORDERED VOCABULARY • NOT SEVERITY • NOT RISK

VERIFIEDDERIVEDPARTIALASSIGNED

DISPUTED / OPENTBDNO EVIDENCE

RESOLVED / NOT-A-CONFLICT — 2 AREA TAXONOMY ACTIVE CONFLICT: NO

1,248 m² ≠ 550 m²
TOTAL USEFUL BUILDING AREA vs COMMON / SHARED BUILDING AREA • CONFLICT: NO

6,403.8 m² ≠ ≈15,020 m²
CADASTRAL BUILDING AREA vs ADJACENT / ASSOCIATED TERRITORY • CONFLICT: NO

PROVENANCE CLAIM → SOURCE → STATUS → CONFIDENCE → VALIDATION ACTION • NO VISUAL REPRESENTATION MAY PROMOTE CONFIDENCE

SUPPORTING RULES • OPEN CONFLICT ≠ OWNER DECISION • TBD LOCATION ≠ NO PHYSICAL COMPONENT • ASSIGNED TAXONOMY ≠ VERIFIED GEOMETRY

OPEN ITEM ≠ DATA ERROR

UNKNOWN ≠ ABSENT

PARTIAL ≠ VERIFIED

NO EVIDENCE ≠ PROVED ABSENCE

VISUAL CONFIDENCE ≠ SOURCE CONFIDENCE

SITE SURVEY & 3D READINESS HANDOFF

A3-EC01-06 · 3D / PILOT READINESS MATRIX

PRODUCTION P2-06 · STAGE P2_CONTEXT_REAL_BUILDING
TYPE PT-06 REFERENCE FLOW / TBD REGISTER
IR FROZEN · PAGE SPEC PAGE_SPEC_READY

PRIMARY ENGINEERING QUESTION

What EC-01 site-survey and MCD-baseline facts are actually usable for 3D massing and pilot work today, and which remain blocked pending field validation?

CORE THESIS

3D MASSING MAY PROCEED ON FLOOR COUNT AND FOOTPRINT SHAPE ALONE.

Nothing else is ready. Unknown engineering geometry remains unknown in 3D.

PRIMARY VISUAL — 3D USABILITY MATRIX (7 rows, source order preserved · 6 of 7 require field confirmation)

FACT	DIRECT USE IN BLENDER?	USE AS CONSTRAINT?	NEEDS FIELD CONFIRMATION?	DO NOT MODEL	PROVENANCE
9 floors	Yes	—	No	—	VERIFIED
Stepped (non-rectangular) footprint outline, Image 3	No — dimensions incomplete	Yes — DERIVED constraint on massing shape, supersedes a simple-rectangle assumption	Yes, before final massing	—	DERIVED
Footprint area (site-coverage)	No — not established	No — 1,248 m² / 550 m² are area sub-categories, not footprint values	Yes — needs dimensioned plan or field survey	Do not model any area-derived footprint	PARTIAL
3 entrance/staircase cores	No exact coordinates	Yes, as section-count constraint	Yes (exact spacing)	—	TBD
Floor plan room layout (SRC-004 photographed sheets)	Partial — legible portions only	Yes — DERIVED reference, one representative floor	Yes, for illegible dimensions	—	PARTIAL
Roof/basement	No	No	Yes	Do not invent	TBD
Engineering rooms/risers	No	No	Yes	Do not invent — no location evidence exists	TBD

SITE-SURVEY / FIELD-CONFIRMATION GAPS (5 categories)

PHYSICAL

Entrance layout/numbering vs. documented layout; basement access; technical-room locations; roof access; footprint outline verification.

ENGINEERING

Heat entry/metering, water entry/meter, electrical main board, sewer collector/outlet locations; gas presence/absence.

DEVICE DEPLOYMENT

Mounting positions, available power, environmental conditions, access restrictions — none currently known.

CONNECTIVITY

Cellular signal conditions, basement/technical-room penetration, candidate edge location, direct-cellular feasibility — TBD WITH BEELINE / SITE SURVEY.

DOCUMENTATION

EVIDENCE / PROVENANCE STATES (5 states)

VERIFIED	Directly usable now.
DERIVED	Explicit constraint derived from evidence, not a direct source fact.
PARTIAL	Some evidence exists; not established.
TBD	Not established; requires field confirmation.
NO EVIDENCE	Reviewed evidence does not support the claim — absence is not proved.

Epistemic evidence axis, not a severity ladder.

REFERENCE CANDIDATE — HEAT-03 (not installed / not validated)

Question: can a real HEAT-03 (or equivalent heating mixing/elevator node) be identified from primary evidence?

NO EVIDENCE

 — answer, as recorded in the frozen source extract

Central heating connection existence confirmed only; no basement plan, mixing-node location, or riser topology found.

Boundary: HEAT-03 and all HEAT-xx candidates remain unassigned ontology candidates, not instantiated facts.

Same fact tracked as CI-03 on A3-EC01-05 — cited here, not re-registered.

FROZEN SEMANTIC COUNTS

7	5	6	5	7
PRIMARY UNITS	SECONDARY UNITS	RELATIONSHIPS	EVIDENCE STATES	OPEN / TBD

VALIDATION & INCLUSION DISCIPLINE

VALIDATION REQUIREMENT COUNT: 6 · GUARDRAIL COUNT: 2
SEM-MCD-001 INCLUDED: YES · HEAT-03 INCLUDED: YES

GUARDRAILS — 2

- UNKNOWN ENGINEERING GEOMETRY REMAINS UNKNOWN IN 3D — THIS PAGE DOES NOT CREATE OR MODIFY ANY 3D GEOMETRY.
- HEAT-03 AND ALL HEAT-xx CANDIDATES REMAIN UNASSIGNED ONTOLOGY CANDIDATES, NOT INSTANTIATED FACTS.

PROHIBITED INFERENCE

No installed device, confirmed location, validated topology, or completed field confirmation.
Missing evidence ≠ absence of an object. Unknown geometry is not shown as known geometry.